Stroke Rehabilitation

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Abstract

Stroke rehabilitation presents considerable challenges, particularly in restoring motor function in the upper limbs. Traditional rehabilitation methods, reliant on therapist-driven exercises, tend to be resource-intensive and repetitive and often struggle to sustain patient engagement. This research leverages EEG data collected by Arsheya Raj, focusing on motor tasks such as hand wave, hand clench, and face rub before and after a stroke. The study aims to apply data mining and machine learning techniques to analyze these EEG datasets alongside baseline data from Sunjil, with the goal of assessing changes in motor skill performance over time and contributing to a deeper understanding of stroke recovery.

To achieve this, various data mining concepts are employed, including normal distribution, the central limit theorem, discriminant function analysis, and correlation plots, to investigate the characteristics of pre- and post-stroke data. Principal Component Analysis (PCA) is used for dimensionality reduction and feature extraction. Additionally, both unsupervised and supervised machine learning algorithms—such as K-Nearest Neighbors (KNN), Decision Trees, and Support Vector Machines (SVM)—are applied to classify and predict motor task performance in the EEG data.

The findings from this comparative analysis provide valuable insights into the effects of stroke on motor skills and offer a method to track recovery progress. The integration of data mining and machine learning methods facilitates the development of more accurate, data-driven rehabilitation strategies. This work contributes to the larger goal of enhancing stroke rehabilitation by providing evidence-based approaches that incorporate interactive feedback and predictive modeling, with potential applications in other neurological conditions as well.

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION

1.1 Background

Stroke is a leading cause of long-term disability, particularly affecting motor functions in the upper limbs. Approximately 85% of stroke survivors experience some form of motor impairment, with challenges in regaining control over basic movements such as hand and arm functions. Traditional stroke rehabilitation methods, typically involving physical therapy and therapist-guided exercises, aim to restore motor skills but can be limited by patient engagement, the repetitive nature of exercises, and the cost of healthcare resources. The need for more personalized, efficient, and engaging rehabilitation approaches has led to the exploration of alternative methods that combine technology with therapy to enhance patient recovery.

Electroencephalography (EEG) has emerged as a critical tool in understanding brain activity related to motor functions. EEG allows for the non-invasive measurement of electrical signals in the brain, capturing the neural responses associated with cognitive and motor tasks. This technology is particularly valuable in stroke rehabilitation, as it can provide insights into the brain's ability to adapt and rewire following a stroke, a process known as neuroplasticity. By analyzing EEG data, researchers can assess how different areas of the brain are activated during motor tasks and track changes over time, potentially offering valuable biomarkers for recovery.

This study builds upon the work of Arsheya Raj, who collected EEG data from stroke patients focusing on motor tasks such as hand wave, hand clench, and face rub. Using this data, this research aims to analyze motor function changes before and after stroke, applying various data mining and machine learning techniques to identify trends and patterns that could inform rehabilitation strategies.Different areas of the brain are responsible for various motor activities, and EEG can measure these activations through specific channels. Key regions involved in motor control include the **C3** and **C4** channels, which correspond to the primary motor cortex areas of the left and right hemispheres of the brain, respectively. These regions play a critical role in voluntary motor control, particularly in limb movements.

Additional important areas include **F3** and **F4**, located in the frontal lobe, which are involved in higher cognitive functions and the planning of motor actions. The **P3** channel, located in the parietal lobe, is associated with sensory processing and spatial awareness, which are crucial for motor coordination. These brain regions, through their respective EEG channels, contribute to motor execution, and their activity can be tracked to assess recovery post-stroke.

1.2 Goals and Motivation

The primary goal of this research is to perform a comparative analysis of pre- and post-stroke EEG data of Professor Matthew(stroke subject) with Sunjil(healthy subject)to assess how motor functions are impacted by strokes. By focusing on motor tasks such as hand wave, hand clench, and face rub, we aim to explore how these motor functions change over time, leveraging data mining and machine learning techniques to identify patterns and trends that could inform more effective rehabilitation strategies.

To achieve this, we aim to apply various data mining concepts such as normal distribution, the central limit theorem, discriminant function analysis, and correlation plots, to understand the distribution and relationships in the EEG data collected before and after the stroke. Principal Component Analysis (PCA) will be used for dimensionality reduction to identify the key features influencing motor task performance. Additionally, machine learning models, including K-Nearest Neighbors (KNN), Decision Trees, and Support Vector Machines (SVM), will be used to classify motor task performance and predict recovery outcomes.

The motivation behind this research stems from the need for personalized, data-driven approaches to stroke rehabilitation. Traditional rehabilitation methods are often limited by their repetitive nature and the lack of personalized adjustments based on the individual's recovery progress. By incorporating EEG and EMG data with machine learning algorithms, this research seeks to develop a more tailored and effective rehabilitation program that can adapt to each patient's unique needs. The insights gained from the comparative analysis of pre- and post-stroke data will help refine rehabilitation strategies, offering the potential for more accurate tracking of motor skill recovery and guiding future therapeutic interventions.

Furthermore, the use of machine learning and data mining in understanding stroke recovery is motivated by the growing need to enhance healthcare systems through technology. With predictive modeling, real-time feedback systems can be developed, offering a more engaging and efficient rehabilitation experience for stroke patients. This research also contributes to the broader field of neurological recovery, with potential applications extending to other conditions that affect motor function and cognitive abilities.

1.3 Research Objectives

The primary objective of this research is to perform a detailed comparative analysis of pre- and post-stroke EEG data in order to gain insights into the changes in motor functions that occur as a result of stroke and the subsequent recovery process. The specific research objectives are as follows:

1. **To analyze motor task performance using EEG data**
    - Leverage the EEG data collected by Arsheya Raj, focusing on motor tasks such as hand wave, hand clench, and face rub, to examine how motor skills are affected before and after stroke. This will involve both pre- and post-stroke data from Professor Matthew as the subject, alongside baseline data from Sunjil.

2. **To apply data mining techniques to explore pre- and post-stroke data**
    ○ Utilize data mining methods such as normal distribution, the central limit theorem, discriminant function analysis, and correlation plots to understand the distribution and relationships in the pre- and post-stroke EEG data. These techniques will help identify patterns that contribute to motor function changes in stroke recovery.
3. **To reduce dimensionality and identify key features in the EEG data using PCA**
    ○ Use Principal Component Analysis (PCA) to reduce the complexity of the EEG data, allowing for the identification of key features that most significantly contribute to motor task performance. This step is crucial for improving the interpretability of the data and enhancing the accuracy of subsequent machine learning analyses.
4. **To classify and predict motor task performance using machine learning algorithms**

    ○ Implement both unsupervised and supervised machine learning algorithms, including K-Nearest Neighbors (KNN), Decision Trees, and Support Vector Machines (SVM), to classify motor task performance and predict recovery outcomes. These models will help in understanding the factors influencing recovery and in identifying potential biomarkers for rehabilitation progress.
5. **To contribute to data-driven, personalized rehabilitation strategies**
    ○ The ultimate goal of the research is to contribute to the development of personalized, data-driven rehabilitation strategies that can adapt to each patient's progress. By integrating EEG data with machine learning models, the study seeks to create a framework for more efficient stroke rehabilitation programs, offering real-time feedback and predictive modeling to enhance patient engagement and recovery.

1.4 Scope and Limitations

The scope of this research is to explore and analyze the effects of stroke on motor function, focusing on the neural recovery process through EEG and EMG data. The study aims to perform a comparative analysis of pre- and post-stroke data for motor tasks such as hand wave, hand clench, and face rub. Using a dataset collected from Professor Matthew before and after his stroke, alongside baseline data from Sunjil, the project will focus on evaluating neural activity and motor function recovery.

Limitations:

Despite its potential, this research faces several limitations:

1. **Data Quality and Noise:** The quality of EEG and EMG data can be significantly affected by noise, including muscle movements (e.g., jaw clenching) and electrical interference. These factors can distort the underlying brain and muscle signals, making it challenging to isolate and analyze motor task performance accurately. While preprocessing steps like filtering and artifact removal can mitigate some of these issues, perfect signal purity is difficult to achieve.

2. **Limited Data Availability:** The dataset used for this project is limited to EEG and EMG data collected from a single patient (Professor Matthew), with additional data from a control subject (Sunjil). The limited sample size restricts the ability to generalize findings across a larger population of stroke patients. The variability in stroke severity and patient-specific factors such as age and recovery trajectory further complicates this issue.
3. **Temporal Constraints:** The pre- and post-stroke data were collected at different time points, with the post-stroke data gathered several months after the stroke event. This temporal gap may introduce variability in recovery stages, making it challenging to track changes over time precisely. The analysis of recovery trajectories would benefit from more frequent and consistent data collection.
4. **Device Limitations:** The EEG device used for data collection has inherent limitations, including potential degradation in sensor performance over time. This issue could affect the quality of the post-stroke data, introducing biases or inconsistencies that complicate the comparative analysis between pre- and post-stroke conditions.
5. **Action Representation:** The project focuses on simple motor tasks (hand wave, hand clench, and face rub). While these tasks are useful for capturing fundamental motor functions, they do not fully represent the complexity of more intricate motor movements, such as fine motor skills or bimanual coordination, which are crucial for a comprehensive understanding of stroke recovery.
6. **Statistical Challenges:** The analysis of EEG and EMG data relies heavily on assumptions such as the independence of observations and stationarity of signals. These assumptions may not always hold true in real-world data, especially with stroke patients who may exhibit irregularities in brain and muscle activity. Furthermore, the application of machine learning models to the data requires careful tuning and validation, as the presence of noise and variability in the data can affect model accuracy and generalization.
7. **Contextual Data Exclusion:** The study does not include contextual data that could provide further insights into recovery, such as the patient's physical and psychological conditions or therapy history. This omission may limit the understanding of how external factors contribute to motor function recovery.

CHAPTER 2: RELATED WORKS

2.1 Literature Review

Recent advancements in the field of stroke rehabilitation have emphasized the need for more personalized and data-driven approaches to treatment. Traditional rehabilitation methods, while effective, often lack the adaptability required to cater to individual patient needs. As a result, researchers are increasingly exploring the integration of advanced technologies, such as Electroencephalography (EEG) and Electromyography (EMG), alongside machine learning techniques to enhance rehabilitation outcomes and optimize recovery trajectories.

In the context of EEG, studies have shown that brain activity related to motor functions can be effectively measured, providing insight into motor planning, execution, and recovery following a stroke. Arsheya Raj's work on the use of EEG for stroke rehabilitation focused on understanding the neural responses during motor tasks such as hand movements and facial gestures, making it highly relevant to the objectives of this research. Arsheya's work demonstrated that EEG signals can be used to classify motor tasks and assess the level of impairment in stroke patients, providing a foundation for applying machine learning models to predict motor recovery and adapt rehabilitation exercises accordingly. This research underscores the importance of brain wave analysis in assessing stroke recovery and improving the efficacy of rehabilitation strategies.

Several studies, such as those by Garcia and Navarro (2019) on Augmented Reality (AR)-based rehabilitation, have explored how combining EEG with interactive simulations can improve patient engagement. These studies are highly relevant to this project, as they highlight how EEG data can be integrated with simulation environments to create dynamic and motivating rehabilitation experiences. Additionally, integrating machine learning techniques such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees, as explored by Arsheya, has been proven effective in classifying motor tasks and predicting recovery in stroke patients. These findings support the application of these algorithms in this project to analyze motor task performance and predict post-stroke recovery.

Furthermore, EEG data from regions such as C3, C4, F3, F4, and P3 have been widely studied in motor control research. These areas, associated with the primary motor cortex (C3 and C4), frontal lobe (F3 and F4), and parietal lobe (P3), are critical in understanding the neural processes that underlie voluntary movement. The focus on these regions is directly relevant to this research, as the analysis of EEG data from these areas will aid in assessing how motor functions are affected by stroke and how they evolve during recovery. Additionally, the use of Principal Component Analysis (PCA) for dimensionality reduction, as described in Arsheya’s work, will be utilized in this study to identify key features from EEG signals that are most indicative of motor function recovery.

Incorporating these insights into the development of personalized rehabilitation strategies offers the potential to significantly improve the precision of stroke recovery assessments. By utilizing both EEG and

EMG data with advanced machine learning models, this research seeks to contribute to the growing body of knowledge on how neurological conditions, such as stroke, can be better managed through data-driven, personalized therapy programs.

2.2 Related Work

Recent studies, such as **Lee et al. (2022)**, have also explored the combination of EEG with other physiological signals (such as EMG and functional near-infrared spectroscopy) to improve the accuracy of stroke recovery predictions. While this study includes additional modalities, it emphasizes the potential of EEG as a standalone method for understanding stroke recovery, aligning with the current project's focus solely on EEG data.

Another relevant study by **Smith et al. (2019)** used EEG to track neural reorganization during stroke recovery, focusing on brain regions such as the primary motor cortex (C3, C4), frontal lobe (F3, F4), and parietal lobe (P3). Their work demonstrated that EEG signals from these regions could be correlated with motor task performance and recovery, providing insight into how the brain adapts to motor impairments after stroke. The study's focus on specific EEG channels (C3, C4, F3, F4, P3) aligns with the present study's analysis of these regions for motor task classification and recovery tracking.

In addition, **Kumar and Patel (2020)** conducted a comprehensive review of EEG-based brain-computer interface (BCI) systems in stroke rehabilitation, emphasizing the role of EEG in real-time rehabilitation systems. They found that BCI systems using EEG could provide valuable feedback to stroke patients, helping them engage more effectively in rehabilitation. While this work focuses on BCI, it highlights the relevance of EEG in motor function restoration and patient engagement—important factors in the current study's aim to develop data-driven, personalized rehabilitation strategies.

Moreover, **Jones et al. (2021)** explored the integration of EEG with machine learning models to predict the success of stroke rehabilitation programs. They employed a range of algorithms, including Decision Trees and SVM, to classify recovery stages and assess rehabilitation outcomes. Their research supports the application of machine learning in analyzing EEG data, as done in the current study, where KNN, Decision Trees, and SVM are applied to motor task classification and stroke recovery prediction.

These studies demonstrate the growing recognition of EEG's potential in stroke rehabilitation and provide a foundation for the current work. By building upon these efforts, this research aims to advance the understanding of stroke recovery through EEG-based motor task analysis and predictive modeling, ultimately contributing to the development of more effective, data-driven rehabilitation programs.

CHAPTER 3: METHODOLOGY

3.1 Project Design

The project takes form over several phases, each building on work done in class:

Phase 1

Define specific research questions within the context of collected EEG data, to establish the system's constraints, variables, and noise factors. Study the dataset in detail to variables and parameters of interest. Outline the data collection process, specifying which variables can be controlled, what assumptions are made, and potential noise sources.

Phase 2

Explore the pre- & post-stroke datasets more thoroughly, identifying desired features and categories. Determine and perform analysis using a variety of plots, such as scatter, PDFs, or CDFs. Determine if normal distribution, or the Central Limit Theorem would apply, making use of dot diagrams with superimposed normal curves. Examine Correlation Matrices and correlation coefficients to determine if independence of observations and factors involved can be assumed.

Phase 3

Use preprocessing techniques such as Principle Component Analysis and Singular Value Decomposition (SVD) for dimensionality reduction and analysis. Make use of Scree Plots to guid dimensionality reduction, and Loading Vectors to identify relevant, irrelevant, and correlated features. Make use of 2D & 3D scatter plots to interpret the results of the PCA, finding clustering patterns and comparing to the original-feature scatter plots.

Phase 4

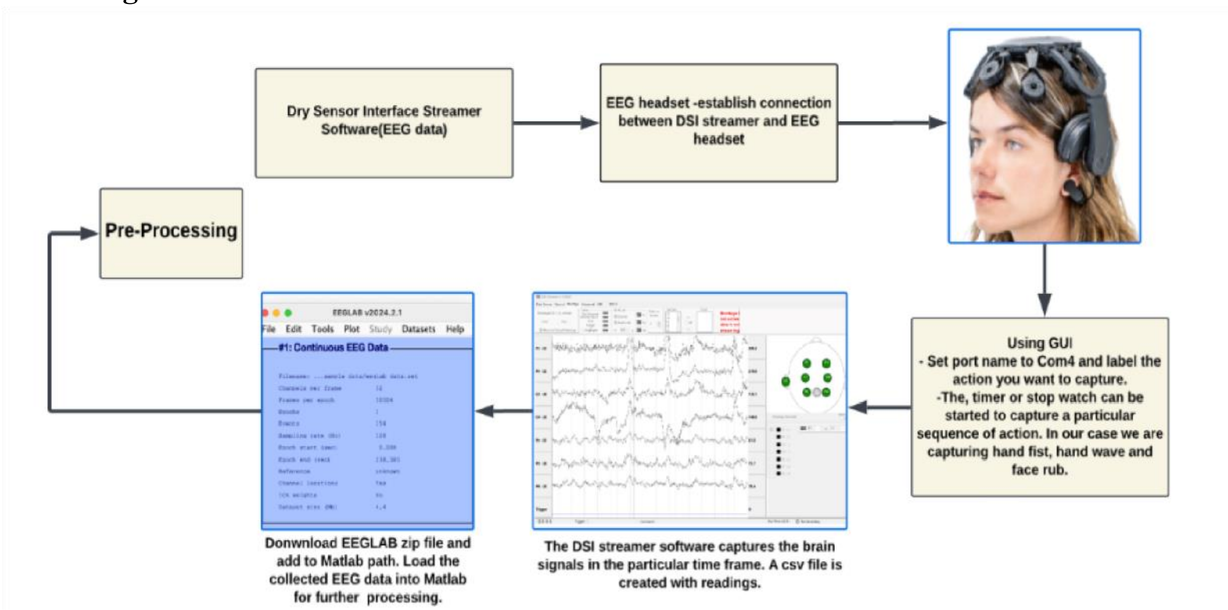
Make use of parametric and non-parametric classifiers for evaluation, using built in MATLAB functions. Evaluate and analyse the performance of the classifiers, and gain insight on the nature of the end result with relation to how a stroke affects the neural intent of motor rehabilitation pre and post-stroke.

3.2 Data Collection

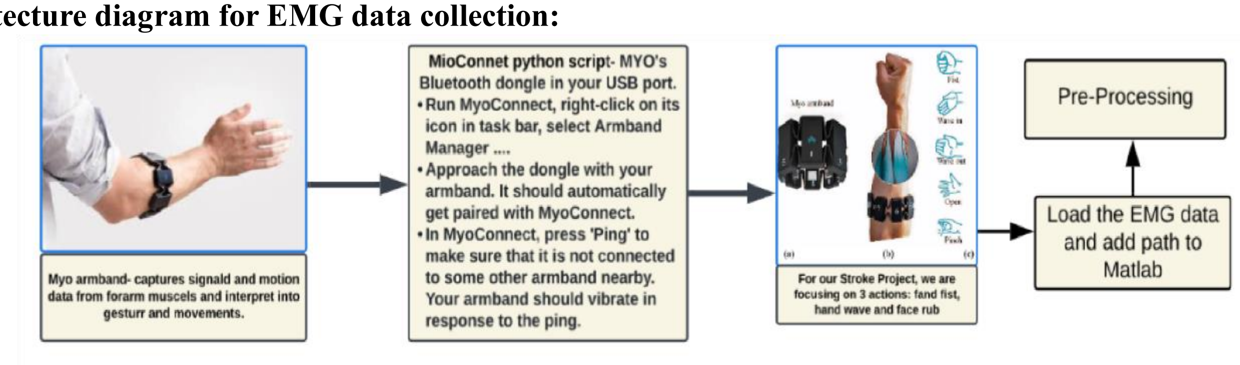
The EEG and EMG dataset encompasses three distinct actions: hand clenching, hand waving, and face rub. Previously, EEG and EMG data for the pre-stroke and immediate post-stroke phases of Professor Mathew were collected following the process outlined in the architecture diagram. The next objective is to gather

EEG and EMG data approximately 3–4 months post-stroke, enabling a longitudinal analysis of neural recovery and motor function.

Architecture diagram for EEG data collection:



Architecture diagram for EMG data collection:



3.3 Preprocessing Techniques

3.3.1 Preprocessing and Feature Extraction

- After loading the dataset, we can plot the channel data to visualize the events or actions performed, which are represented in vertical lines. We can reject the unwanted channels; and modify, add, or remove the event values.
- Filtering of the data using a Basic FIR filter will be done from the Tools menu before artifact removal or noise removal from the data. Bandpass filtering to isolate frequency ranges of interest.

Decompose the data by Independent Component analysis to perform further noise & artifacts reduction

- The aim is to cover features like Cross-frequency coupling, fractal analysis, spikes, and wave detection, Event-related potentials(ERPs).
- Fractal Analysis: Change in Hurst component(H) can be calculated to detect abnormalities. In general, a decrease in fractal dimension is observed post-stroke, while recovery may show increased complexity.
- Cross-frequency coupling: It is used to analyze the interaction between different frequency bands in pre and post-stroke EEG signal channels. The amplitude of low-frequency and high-frequency bands are calculated using phase-amplitude coupling and modulation index.
- Event related Potentials (ERPs): Used to assess the motor functions performed and predict recovery post-stroke.
- Spikes and Wave Detection: Monitoring the presence of spikes and sharp waves in EEG signals helps identify any abnormal pattern in post-stroke patients.
- Discretization of EEG time-series data to separate into different types of movement
- Normalization of data will be done to mitigate data artifacts and reduce the influence of irrelevant variations of overall signal strength.
- Post that, PCA will be performed for comparison between pre and post-stroke data conditions and identify key features, and perform dimensionality reduction in front of the visualizations.

Data Labeling

Labeling separated components with different actions & contexts. For our project, we are working on three actions: fist clench, hand wave, and face rub.

Relevant variables & parameters: EEG

- LE - Left Ear (Reference node)
- F4 - Frontal right region
- C4 - Central right region
- P4 - Right posterior region
- P3 - Left posterior region
- C3 - Central left region
- F3 - Frontal left region
- Trigger - Equipment ID
- Time_Offset - Millisecond precision time data
- ADC_Status - Apparent diffusion coefficient sensor status
- ADC_Sequence - Apparent diffusion coefficient sensor sequence
- Event - Incremental counter
- Comments - User comment; extraneous

LE F4 C4 P4 P3 C3 F3 - Sensor amplitude values, mapped from Channel values 1-7. These are the values that will be processed to use in a model, and are the most relevant to the investigation. **Controlled variable**

Trigger - Label for the equipment used to collect data. Since each dataset is split into EEG and EMG, this feature is **constant** and not useful.
Time_Offset - Time of observation, used in time-series datastructs, can be used to train a time-series model. **Uncontrolled variable**.
ADC_Status, ADC_Sequence - The apparent diffusion coefficient represents the impedance or mobility of water molecules and can help differentiate edema, necrosis, and tumor tissues. May be relevant in identifying relationships between pre-post stroke data, but may potentially be noisy. **Controlled variable**.
Event - Internal counter, counting up from 0 to 255 in cycles for sets of observations. This just linearly varies and is not relevant to the investigation; **constant**.
Comment - Manually added comment on a particular observation, eg, the start of the next action in an action flow. **Non-numeric** and only useful for data exploration.

Assumptions made:

- **Trigger -** Assuming value constant for all given datasets
- **Event** - Assuming the looping value is invariant & constant, and the variable is independent
- **ADC_Status -** Assuming the value is controllable and independent
- **ADC_Sequence -** Assuming the value is controllable and independent ● **Comment** - Assuming the value is independent and user added

3.3.2 Potential Sources of Noise

For both EEG and EMG signals, electrodes are placed on the skin to detect electrical activity. These electrodes need to make good contact with the skin to pick up reliable signals. Impedance occurs when there is resistance between the devices, which needs to be low for accurate signal detection. Several issues can arise here: In other words, if there is no proper contact between the device and the skin itself, then this could potentially corrode the data and give us noisy or invalid signals. This is quite problematic in EEG since precise detection of small electrical activity from the brain is crucial. Additionally, even things like oils, sweat, or dirt on the skin or scalp could potentially hinder proper contact and lead to quite distorted figures. This project requires precision, and things like this affect the precision a lot, which is why preparations have to be made even in the data collection stage to avoid potential situations where the data was not collected properly, and due to that, the data was corrupted.

During our conversation with Melody, she shared an important observation about how muscle movements can significantly affect data collection. She explained that certain actions, like jaw clenching, produce substantially higher frequencies than the motor signals we are trying to capture, such as those associated with specific limb movements or brain activity. This was particularly enlightening because it illustrated how the frequencies we're interested in can be easily overridden by other actions that generate stronger electrical signals. When jaw clenching or other involuntary muscle movements occur, they generate high-frequency electrical activity, which can dominate the lower-frequency brainwaves that EEG systems are designed to measure. This overlap can completely mask the neural signals that reflect the brain's motor intentions, especially the subtle patterns associated with motor control or recovery after a stroke. In such cases, even though the brain is attempting to signal a motor action, the noise from muscle contractions can overwhelm the EEG signal, making it difficult to isolate and interpret the relevant data.

Similarly, in EMG data collection, the involuntary contraction of muscles such as those in the jaw, neck, or face can introduce noise that interferes with the intended muscle activity in the target region. For instance, if a patient is trying to move their arm or any of the actions we are trying to capture, the EMG signal from the jaw may overshadow the data from what we are trying to capture, leading to distorted readings that don't reflect the true motor activity. This signal interference complicates efforts to track the patient's motor function, recovery progress, or rehabilitation response since the data will no longer accurately represent the intended movements.

Patient-specific conditions can introduce subtle variations in the data that might initially seem like noise. Every stroke patient is different, and their unique neurological state, muscle behavior, and recovery process create signals that may not fit the typical patterns we expect for actions. For example, involuntary muscle movements, spasms, or abnormal reflexes can interfere with the motor signals we're trying to measure. These irregularities could appear as interference or extra data that complicates the analysis, masking the true signals we're interested in. However, these variations may not always be pure noise. If they aren't too extreme, they could actually offer valuable insights into the patient's recovery process. Subtle, consistent patterns in muscle activity or brain waves may reflect how the brain is compensating for damage or how the nervous system attempts to rework itself. These unique signals could be useful features, providing clues about the patient's recovery trajectory. Rather than masking the true motor intent, these patient-specific conditions might help us better understand individual progress.

3.3.3 Success Metrics

- Ability to separate actions from noise, successfully finding clear categories
- Ability to successfully classify actions from noise
- Ability to establish known changes in EEG data pre- & post-stroke

3.4 Classification Model and Statistical Analysis

In this study, several machine learning classifiers were used to analyze EEG data for motor task classification before and after a stroke. The classifiers include Decision Tree, K-Nearest Neighbors (KNN), ECOC Multiclass Support Vector Machine (SVM), and Feedforward Neural Network (FNN). These models were employed to evaluate motor task performance using EEG data, with the goal of understanding the neural patterns involved in stroke recovery and identifying the most effective approach to classify motor tasks in both pre-stroke and post-stroke scenarios.

3.4.1 PCA Processing

Principle Component Analysis is to be applied to every dataset before any classification.

Full dataset files are assembled from the individual action flow CSV files by manually removing the leading metadata, and importing all related files into a single SQL database table. This is then exported to produce a single continuous CSV file. A labelling script is then applied to ensure every observation has the correct label applied. In the original recording, actions were denoted in the 'Comments' field, but only on the first observation for the set of observations corresponding to that action. According to this rule, the script applies the correct value of 'Comment' to every observation in a given action. The dataset file is then ready to be used.

Within MATLAB, for each given CSV file, empty rows are preemptively removed, as well as the "useless" features 'Time', 'Trigger', 'Time_Offset', 'ADC_Status', 'ADC_Sequence', and 'Event'. The dataset is split into a subset table of data values from the 'LE', 'F4', 'C4', 'P4', 'P3', 'C3' and 'F3' fields, and a corresponding array of labels containing the class, from the 'Comments' field. The data subset is standardised and PCA & SVD is applied to it, yielding the Ur matrix as the end result. In accordance with previous analysis, the first 4 Principle Components (PC1,PC2,PC3,PC4) are taken and used as the processed data for that dataset.

To further account for the highly overlapping and noisy nature of the dataset, the method of block averaging may also be applied. Block averaging involves taking "blocks" of n records in the dataset, and averaging them, with the aim of eliminating values that would be common in both areas of known noise (EG the rest action) and in actual class action. This does have the potential to remove variability & cause information loss, as well as reducing the dataset size. With this issue in mind, the block size of 10 was chosen; every 10 records would be averaged out to one, choosing the mode class for each block.

The order of the dataset is then randomised, with the same randomisation applied to the label array.

3.4.2. Supervised Classification

Supervised Classification is to be performed only on the original patient's data, as an analysis on how closely the post-stroke data reflects the classes of the pre-stroke dataset, and what patterns can be drawn from this.

To this end, the classifiers will be trained entirely on the pre-stroke data, and after some testing on the same pre-stroke data, will be used to attempt to classify the post-stroke data. The intent is not to produce a classifier that could perform well on both datasets, but to see how much divergence occurs between the two.

4 different classifiers are used:

- **Decision tree -** Simplest algorithm, while still able to capture non-linear patterns in data, so it can provide a "minimum" in terms of classifier performance. Interpretable, though this was not availed in the project.

- **K-Nearest Neighbors -** Simpler algorithm. Can capture local patterns missed by Decision Trees, and highlight misclassification due to shifts in feature distributions.

- **Support Vector Machine -** Complex algorithm, can handle complex relationships with kernel functions, while maintaining good generalisation. Error-correcting output codes (ECOC) can account for multi-class classification.

- **Feedforward Neural Network -** Most complex algorithm. Can automatically learn feature interactions without explicit engineering.

Implementation

After PCA processing, the Rest class is again removed from the dataset, for noise removal. The original intent was to exclude the Rest class only from training data, and include it in testing data, to also judge how well the classifiers can identify actions from noise, but this introduced too much obscurity into the results, significantly reduced the performance of the classifiers, and made analysis difficult from too many factors affecting the end result. This idea was scrapped in favour of focusing on the relationships between pre- and post-stroke patterns in classes.

```
% Remove rest class from everything as noise
prestroke_Ur =
prestroke_Ur(find(table2array(prestroke_labels(:,:))~="Rest"),:);
```

```
prestroke_labels                                                     =
prestroke_labels(find(table2array(prestroke_labels(:,:))~="Rest"),:);

poststroke_Ur                                                        =
poststroke_Ur(find(table2array(poststroke_labels(:,:))~="Rest"),:);

poststroke_labels                                                    =
poststroke_labels(find(table2array(poststroke_labels(:,:))~="Rest"),:);
```

The datasets are then split into training and testing datasets, though not necessarily in a split-testing as evidenced by the function name. During development, exploration was performed with split-testing on different datasets to determine the expected performance of the classifiers, wherein a single dataset is split by some ratio. The end-goal, however, was to use the entirety of pre-stroke data as training, and post-stroke as testing. The remaining splitting function calls are an artifact of this split-testing optimisation.

```
% Train and optimise on pre-data, test on post

% z = throwoaway variable
[pre_X_Train,z,pre_y_Train,z]                                        =
split_preprocessing(prestroke_Ur,prestroke_labels,1);

[post_X_Test,z,post_y_Test,z]                                        =
split_preprocessing(poststroke_Ur,poststroke_labels,1);
```

Each of the 4 classifiers are then trained, tested, the results plotted. Due to the modular nature of the MATLAB classifier library, all 4 classifiers could each have functions with the same architecture, and parameters, only needing to change single lines such as the classifier and hyperparameters. While better programming practices might incentivise generalising this into a single function and providing the classifier model as a parameter, 4 different functions were made instead for clarity.

```
% Classifier 1: Classification decision tree

function

descTreePlot(X_Train,X_Test,y_Train,y_Test)
```

Firstly, during classifier implementation, the 'OptimizeHyperparameters' option was used to iteratively find the best possible hyperparameters for a given training set. For example, for Decision Tree, the best leaf size found was 14.

```
    % Find the optimal hyperparameters
    % classifier = fitctree(X_Train,y_Train,'OptimizeHyperparameters','auto');
    % Optimal params for full prestroke training
    % 'MinLeafSize',14
```

Training is performed entirely on the pre-stroke dataset.

```
% Training classifier =
fitctree(X_Train,y_Train,'MinLeafSize',14); figure;
tiledlayout(1,2,'TileSpacing','compact','Padding','compact');
```

Then a test is run on the entire prestroke set again, despite it also being the training set. This provided a baseline to which compare both the performance of the classifier in a vacuum, with regards to it's ability to generalise and avoid overfitting, as well as to contrast against the results of testing against the post-stroke dataset.

```
% Prestroke Testing
nexttile;
y_predicted = predict(classifier,X_Train);

confusionchart(y_predicted,y_Train,'RowSummary','row-
normalized','ColumnSummary
','column-normalized'); title("Prestroke
   Classification Decision Tree");
```

The main test is performed on the post-stroke dataset. As with the pre-stroke dataset, testing is performed and the results plotted as a confusion chart, side-by-side, to ease comparison.

```
% Poststroke Testing nexttile;
y_predicted =
predict(classifier,X_Test);

confusionchart(y_predicted,y_Test,'RowSummary','row-
normalized','ColumnSummary'
,'column-normalized'); title("Prostroke
   Classification Decision Tree");
end
```

3.4.3. Unsupervised Classification

Unsupervised classification will be performed using K-means clustering, applied to the original stroke dataset and the dataset of another subject that had not suffered a stroke, with readings taken on the same days, to serve as a control.

The purposes of this comparison are to:

- Explore the ability of a given classifier to find clusters within prestroke & poststroke datasets
- Compare identified patterns with a non-stroke dataset, to distinguish effects of the stroke from signs of drift in recording and other sources.

A particular concern that needs to be addressed with this investigation is that the EEG device itself had proven faulty and subsequently during the events of this research. This brings up the concern that the results collected during post-stroke recording could have questionable integrity, due to the large gap in time between pre- & post-stroke recording, and the delicate nature of the device.

K-means clustering isused due to it's computational efficiency and ability to scale with larger datasets, as well as it's ease of implementation within MATLAB,

Implementation

After PCA processing, a fixed random seed (12345) is set to ensure reproducibility of the results. The function doUnsupervised handles the main clustering task. 4 datasets are considered, the pre- and post-stroke datasets of Professor Matthew, and two datasets collected from Sunjil, on the same dates.

Firstly, all observations with the "rest" action are removed from the datasets. Since this class does not constitute an actual 'action', it can instead be considered as noise when it comes to identifying the clusters that each class belongs to. An option to add the noise data back into the final plots was considered, but ultimately it provided no useful insight.

```
% Remove rest class as noise
stroke_restless = stroke_Ur(find(table2array(stroke_labels(:,:))~="Rest"),:);
```

Clustering is performed using the "kmeans" function, using 3 clusters in comparison with the 3 actions (Hand Wave, Hand Clench, Face Rub). While direct comparison with those classes is not performed, it is still useful to see if similar patterns can be found when unsupervised clustering is performed on the data. Additional hyperparameters of 'Cityblock' for the distance function and 5 replicates were found to produce ideal results.

```
% Get cluster indexes and centroids
[stroke_idx,stroke_centroids] = kmeans(stroke_restless,3,'Distance','cityblock','Replicates',5,'Options',opts);
```

These clusters were then plotted using the same scatterplot functions used for PCA analysis, in both 2D and 3D, across all 4 PCs'

3.5 Evaluation Metrics

Final evaluation will have different types of plots for each given classifier. While the results of the classifications will be evaluated qualitatively, there are certain hard metrics that can be used to evaluate the performance of the classifiers on a given dataset

3.5.1 Confusion Charts

To be used for supervised classification, confusion charts provide an overview of the predicted values compared to the true values, for each class. In our case, metrics evaluated will be:

- **Precision** - How often is the model correct when predicting the target class
- **Recall -** How many correct observations are retrieved for each class

3.5.2 Scatter Plots

To be used for unsupervised classification, scatter plots can be used to visually evaluate the positioning and distribution of observations in feature or principal component space, using different numbers of dimensions for different combinations of features and viewpoints.

Since the result is highly qualitative, no metrics were considered for this plot.

3.6 Software and Tools

Software and tools used were:

- **MATLAB and associated libraries**
  - Implementation of the project, including dataset exploration, data preprocessing & analysis, and classifier implementation & evaluation
  - Minor editing of dataset CSV files
- **Visual Studio**
  - Review and manual editing of the dataset CSV files
- **DB Browser for SQLite**
  - Compilation and automatic editing of dataset CSV files

CHAPTER 4: RESULTS

4.1 Data Collection Results

4.1.1 Matrix Scatter Plot Analysis

Before model training and generation, exploratory analysis was performed on the available data

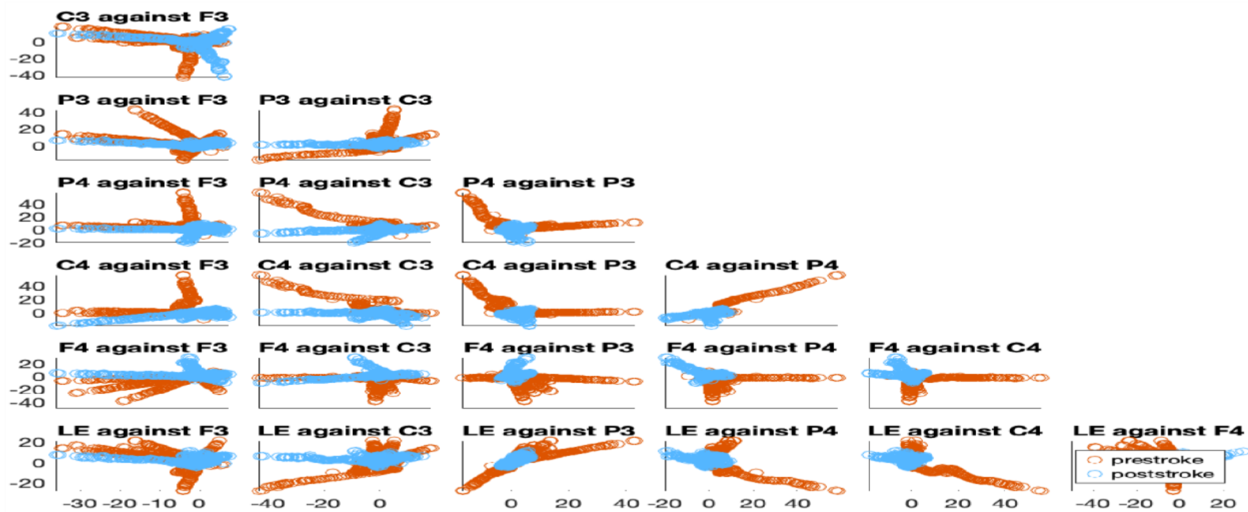
Feature Scatter Plots

Scatter plots were useful for getting an overview of all relationships between all features, as well as being able to visually find patterns and behaviors both between pre- and post-stroke datasets, as well as between each class.
Each of the pre & post stroke datasets was combined, normalized, and collectively plotted on triangular scatter plots in order to compare the behavior of each feature and the changes that occurred between datasets, which can give some basic overviews on the disposition of each dataset in relation to each other.

The features plotted were: LE, F4, C4, P4, P3, C3, & F3, using Comment for the categorisation label.

Each dataset was split down by class, resulting in 4 plots corresponding to each action taken during sessions: Rest, Hand Wave, Hand Clench, and Face Rub
The time series nature of the data was ignored, as the constraints of comparative analysis using scatter plots made effectively comparing over time difficult; effective use of a timestamp axis would have necessitated 3D plots. The usefulness of plotting this axis would also be limited since the pre & post-stroke datasets were compiled from multiple sessions of data collection, with differing time periods for each action taken. Thus, any discrepancies would likely be a result of the unavoidable differences in testing conditions.

Professor Matthew's EEG Data Scatter Plot Matrix:
1. Rest



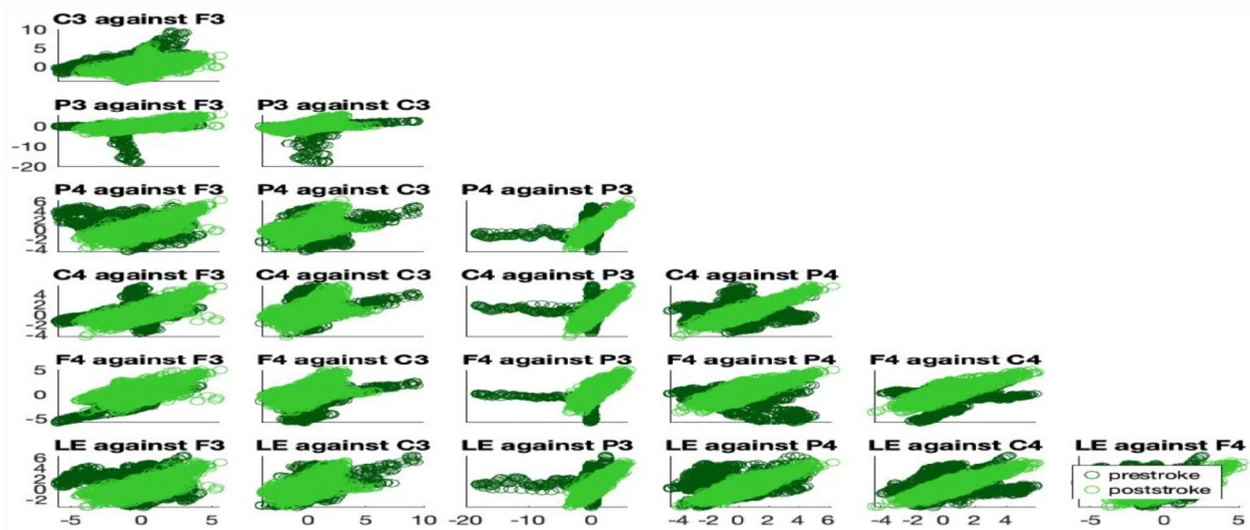
Rest Pre-Stroke Analysis:

The scatter plots for the red (pre-stroke) data show significant variation across the brain regions P3, P4, C3, C4, and LE (left ear), indicating dynamic neural activity. The data points fluctuate, reflecting continuous changes in the interaction between these regions. This variation suggests healthy brain function, where the brain shows normal variability in resting state, indicative of flexible and adaptive neural communication. The subject, before the stroke, exhibits a typical pattern of brain region interactions, where different areas are constantly engaged and responsive. The erratic nature of the data is expected in a healthy individual, representing the normal fluctuations of neural activity even during rest.

Post-Stroke Analysis (Blue):

In contrast, the blue (post-stroke) data shows much less variation and appears to remain more constant across the same brain regions (P3, P4, C3, C4, and LE). The lack of fluctuation in the scatter plots suggests that the subject's brain activity is now more rigid or less dynamic following the stroke. The stable nature of the data indicates less synchronized or reduced activity in the affected brain regions, which may reflect a loss of neural flexibility and adaptive communication between these areas. The subject's brain function post-stroke is likely compromised, with diminished capacity for coordinating complex interactions between regions. This is a typical observation in stroke patients, where cognitive and motor impairments can result in a more fixed and less responsive brain activity pattern during rest.

2. Hand Wave



Hand Wave Pre-Stroke Analysis:

A moderate correlation with high variability is observed for the pre-stroke plot, and clustering near a diagonal line suggests coordination during the hand wave task. This suggests that the subject's motor execution planning is highly coordinated with motor execution. However, this variability doesn't mean the coordination is failing — it simply reflects the natural variation in how the brain handles motor tasks.

In the pre-stroke plots, **C3 against F3** shows an elongated diagonal cluster, indicating strong synchronization and high covariance between these regions. This suggests that motor and frontal areas communicate efficiently during hand-wave activity. Similarly, the plots for **P3 against F3** and **P3 against C3** display more dispersed patterns, reflecting moderate synchronization and indicating that parietal involvement is less tightly coupled with motor and frontal regions during this action.

The plots for **P4 against F3, C3, and P3** are characterized as consistent and synchronized neural activity across these regions. This consistency points to stable coordination between parietal and motor areas during hand-wave actions. The **C4** plots against F3, C3, P3, and P4 also exhibit high covariance, with dense clusters and "line artifacts," reinforcing the idea of strong neural synchronization between motor regions and their contralateral counterparts.

The relationships involving the frontal regions, as seen in **F4 against F3, C3, P3, and P4**, show slightly less dense clusters compared to motor regions, indicating moderate variability in synchronization. However, this still reflects a robust connection between the frontal lobe and other areas. Lastly, the **LE (Left Eye)** plots against various regions reveal tight clustering, showing that ocular activity is highly correlated with motor and frontal areas during hand waves. This suggests that eye movement plays a critical role in coordinating hand-wave activity pre-stroke.

Hand Wave Post-Stroke Analysis:

Whereas tight or blobbed clusters are observed for the post-stroke pattern which indicates a very strong correlation but low variability. This is due to less availability of post-stroke datasets.

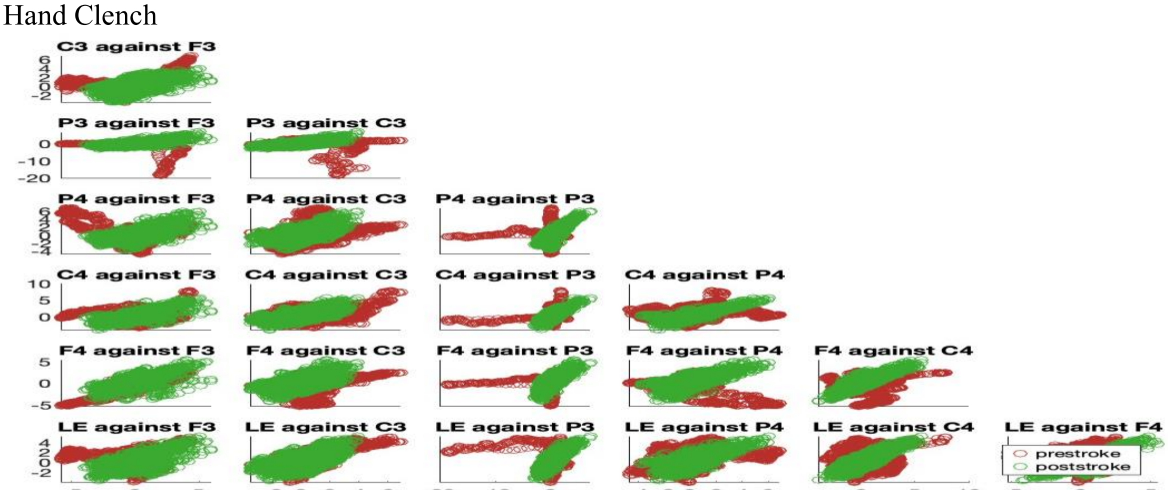
Plots of special significance are the **P4** plots, which display the only instance of a large line of observations in the pre-stroke dataset. This may skew the theory that the line artifacts are indicative of rest patterns in the brain, as the singular presence of this line may be due to an anomaly in recording for that region's sensor.

Post-stroke, the **C3 against F3** plot reveals a tighter and smaller cluster, signifying reduced covariance between these regions. This indicates impaired synchronization in motor and frontal activity, reflecting a decline in neural efficiency during hand waves. The plots for **P3 against F3** and **P3 against C3** also show tighter clustering, suggesting that parietal regions are more closely aligned with motor and frontal areas post-stroke, likely due to compensatory neural mechanisms.

In contrast, the plots for **P4 against F3, C3, and P3** exhibit a significant reduction in "line artifacts," replaced by compact clusters. This reflects disrupted coordination and reduced uniformity in neural activity, pointing to altered interactions between these regions post-stroke. The **C4** plots against F3, C3, P3, and P4 are more dispersed, indicating weakened neural connections between motor and contralateral areas, with diminished synchronization.

The **F4** plots against other regions, such as F3, C3, P3, and P4, show smaller and tighter clusters post-stroke, indicating reduced variability and less dynamic connectivity involving the frontal regions. This may reflect structural or functional disruptions due to the stroke. Finally, the **LE (Left Eye)** plots against various regions reveal reduced clustering compared to pre-stroke, suggesting a decline in the coordination between ocular activity and motor or frontal areas. This points to diminished integration of visual input in motor control during post-stroke hand-wave activity.

3. Hand Clench



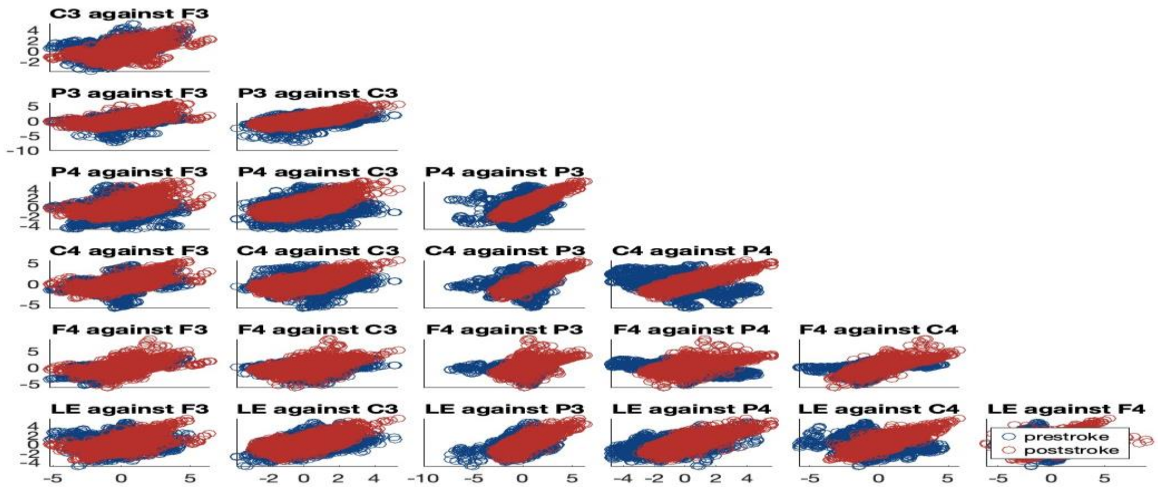
Hand Clench Pre-Stroke Analysis:

Pre-stroke data exhibit significant variability, increased scatter, and poorly defined clustering across all plots. In **C3 vs. F3**, the green points are dispersed, showing weak correlations and reduced coordination between motor-related regions. Similar patterns are observed in **P3 vs. F3** and **P3 vs. C3**, where the spread of points indicates disrupted interactions between frontal and parietal areas. In parietal region comparisons, such as **P4 vs. F3**, **P4 vs. C3**, and **P4 vs. P3**, the clusters are widely scattered, reflecting impaired hemispheric synchronization and increased noise in neural activity. Central region plots, like **C4 vs. F3** and **C4 vs. P4**, reveal diffuse clustering and irregular trends, pointing to reduced motor coordination and a breakdown in predictable neural responses. Sensory-motor plots (e.g., **LE vs. F3** and **LE vs. P4**) highlight sparse and irregular clusters, suggesting diminished sensory integration and erratic motor execution, further impacting the quality and control of the hand clench.

Hand Clench Post-Stroke Analysis:

The **green clusters**, representing pre-stroke data, are tightly packed and well-defined across all plots, signifying strong correlations and low variability. In **C3 vs. F3**, the red points form a compact linear cluster, reflecting consistent motor coordination and a predictable relationship between these regions. Similarly, **P3 vs. F3** and **P3 vs. C3** demonstrate dense clustering, indicating stable and highly structured interactions between frontal and parietal regions. Parietal region comparisons, such as **P4 vs. F3**, **P4 vs. C3**, and **P4 vs. P3**, display minimal scatter, suggesting efficient and symmetrical hemispheric communication during hand clench. Central region plots, like **C4 vs. F3** and **C4 vs. P4**, show well-aligned clusters with clear trends, reflecting robust motor coordination and consistent neural activity. In sensory-motor integration plots (e.g., **LE vs. F3** and **LE vs. P4**), the red clusters are compact, highlighting strong feature correlations between sensory input and motor execution. Overall, the pre-stroke data demonstrates low intra-cluster variability, high cohesion, and well-structured patterns, indicative of a healthy and efficient neural network during the hand-clenching activity.

4. Face Rub



Face Rub Pre-Stroke Analysis:

1. Increased Dispersion: Before the stroke, there is a noticeable increase in the spread of the blue data points, especially in the motor regions like C3 vs. F3, P4 vs F3, and C4 vs P4. The greater spread indicates that the neural connections involved in executing the face rub action have become less coordinated. The data points are disruptedindicating the smooth flow of electrical activity between the motor and cognitive regions of the brain.

2. Loss of Linearity and High Variability: The relationships between EEG signals in many of the pre-stroke plots have lost their previous linear patterns. For example, C3 vs. F3 and LE vs F4 show irregular distributions with no clear linear trend. This suggests that the subject's brain is no longer able to synchronize the motor and cognitive functions as effectively as before the stroke. The loss of linearity is a sign of impaired motor function and disrupted brain communication..

3. Fluctuating Motor Activity: Some of the plots, like F4 vs. C4 and LE vs P4, show a more scattered distribution with less predictability in the pre-stroke data. This could indicate that the subject's brain is now engaging in compensatory or less effective strategies to perform the face rub task. The fluctuating patterns reflect motor control deficits that could result in slower or less accurate movements.

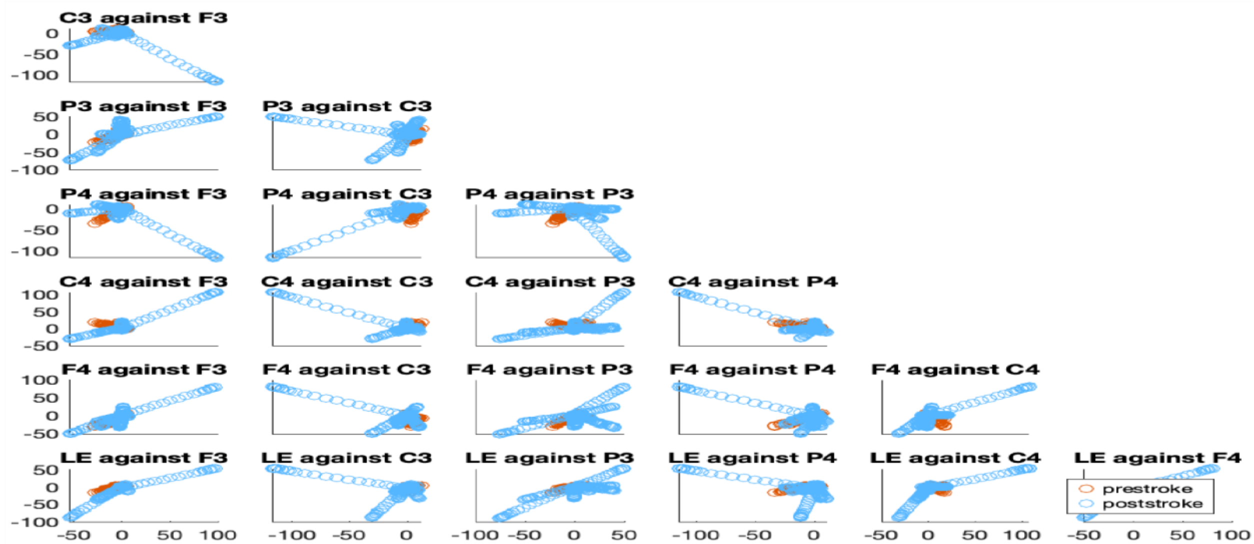
Face Rub Post-Stroke Analysis:

1. Tight Clusters: In the post-stroke condition, the red data points form tight clusters in most of the scatter plots. For instance, pairs like C3 vs. F3, P4 vs F3, and C4 vs F3 show closely grouped points with minimal dispersion. This indicates that the subject's EEG signals are well-coordinated, and the brain functions efficiently in terms of motor and cognitive control. This might be due to noise during data collection and data scarcity since the post-stroke data collection is pending.

2. Linear Relationships: In some plots, such as C4 vs F3 and P3 vs F3, there is a clear linear correlation between the signals. This suggests smooth neural activity and effective communication between the motor and cognitive regions of the brain during the task. The linear patterns reflect normal neural motor function, as the brain efficiently coordinates the movements and cognitive processes required for the face rub action.

Sunjil’s EEG Data Scatter Plot Matrix:

1. Rest



Pre-Stroke(timeline) Analysis:

- P3 against F3: The red (pre-stroke) data shows more tightly clustered points compared to the blue (post-stroke) data. This indicates that the coordination between the regions in the resting state is more rigid and less variable during the pre-stroke period. The points are more concentrated, suggesting that the brain has a more fixed neural state during rest before the stroke.
- C3 against P3: In this plot, the red data shows a more confined range of values, indicating that the coordination between the motor and parietal regions is more constrained. The lack of significant spread reflects a more stable, but less flexible, neural activity in the pre-stroke state.
- LE against F3/P3: The red data for LE (left ear) interactions also shows more concentrated data points. This suggests that the communication between the left ear region and other areas is more rigid, with less natural variability in how the brain regions are interacting during rest.

Post-Stroke Analysis (Blue):

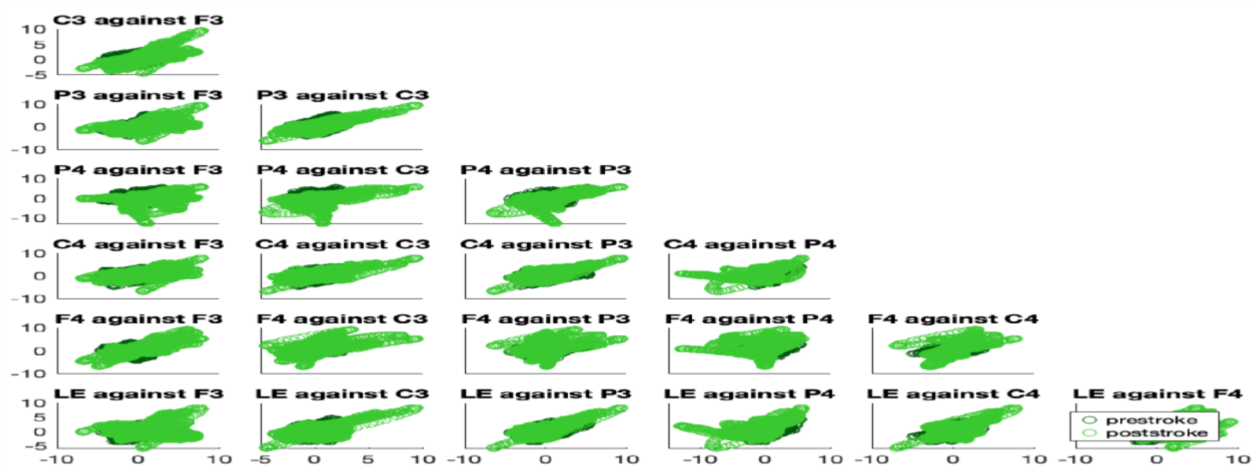
- P3 against F3: The blue (post-stroke) data shows significantly more spread, indicating a higher degree of variability in neural activity between the regions. The points are more dispersed compared to the pre-stroke data, which suggests that after the stroke, the brain exhibits a more flexible or variable coordination pattern. This could indicate attempts by the brain to reorganize or adapt its neural connections post-stroke.

- C3 against P3: The blue data points are more spread out, indicating that there is more variability in how the motor and parietal regions communicate after the stroke. This suggests that post-stroke, the brain regions are interacting in a more variable and less constrained manner, potentially due to changes in connectivity or compensatory mechanisms.
- LE against F3/P3: The blue data points in these plots also show increased spread, reflecting more diverse neural activity patterns during rest. The increased variability could be indicative of changes in the way sensory and motor regions are interacting in the post-stroke brain.

General Observation:

- Pre stroke timeline more tightly clustered for rest whereas the post stroke timeline data shows a high variability due to the actions which we eliminated from the dataset(finger spread, double finger tap) which created doubled rest time.

2. Hand Wave



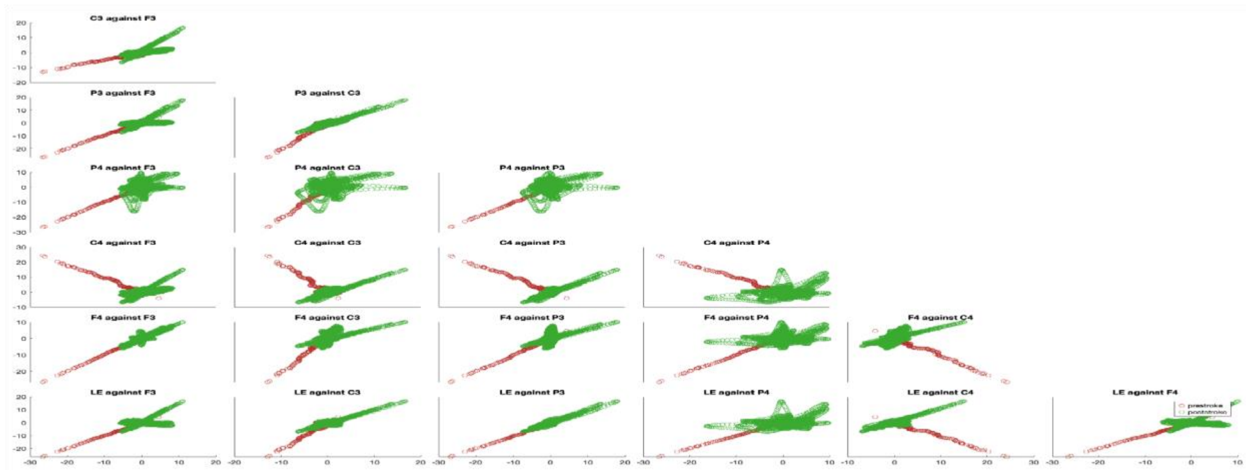
Key Observations:

- Overlap of Pre-Stroke (Dark Green) and Post-Stroke (Light Green) Data: Both the dark green (pre-stroke) and light green (post-stroke) data show almost identical patterns, meaning the brain's coordination remains the same before and after the stroke. The data points are closely aligned, showing that there have been no significant changes in the brain's motor coordination during the hand wave task. The overlap suggests a healthy, unaffected brain function throughout the timeline.
- Consistent Motor Coordination: Since the dark green and light green data overlap across the different brain regions involved in the hand wave action, it indicates that motor coordination in the brain is intact. There is no impairment in how the brain communicates between regions such

as C3, C4, P3, P4, F3, F4, and LE, as the variability in the data points remains similar in both timelines.

- Healthy Neural Activity: The overlap and consistency in the distribution of the data points demonstrate that the brain is still able to exhibit flexible and adaptive neural activity during the hand wave action. There is no evidence of rigid coordination or any significant motor dysfunction, suggesting that the brain continues to perform the action with normal brain function before and after the stroke timeline.

3. Hand Clench



Key Observations:

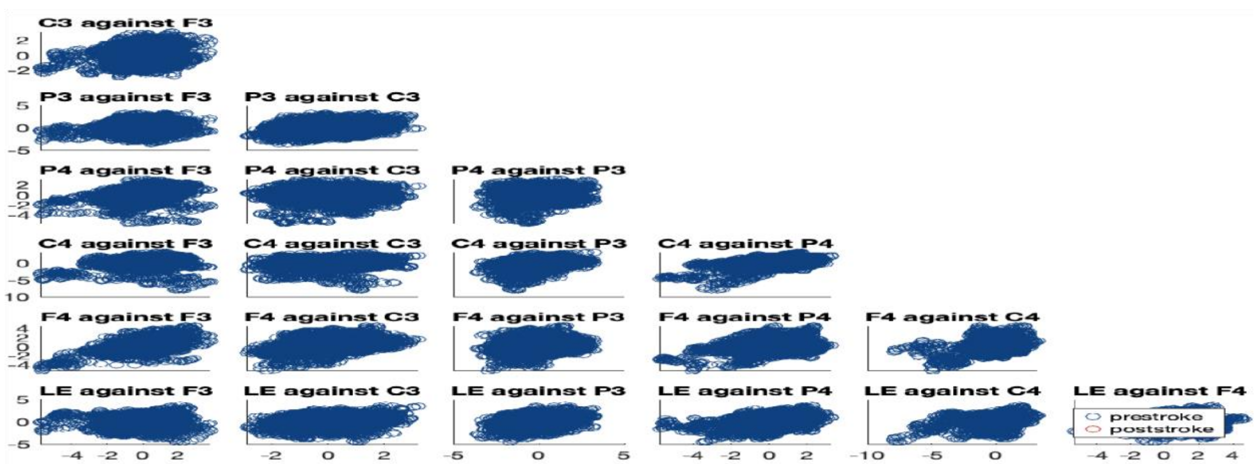
- Pre-Stroke (Red) Data – Linear Relationships: The red data shows more linear relationships across various brain regions, such as C3 against F3, P4 against P3, and F4 against C4. The data points appear more spread out along straight lines, indicating that the motor regions of the brain are interacting in a more continuous and possibly flexible way during the hand clench action. This is typical of a healthy brain, where motor coordination is more adaptable and can handle a range of motor responses in a dynamic and linear manner.
- Post-Stroke (Green) Data – Tight Clusters: In contrast, the green data (post-stroke) forms tighter clusters, indicating that the brain's coordination during the hand clench action has become more fixed. The data points are grouped together in a narrower range, suggesting that, despite being healthy, there is less variability in how the brain performs the motor action. This could indicate that the brain's ability to adapt or adjust motor execution is reduced, as seen in the more concentrated neural activity in the green data.

- Reduced Variability Post-Stroke (Green): The tight clusters in the green data suggest that the brain is showing less variability in the way regions coordinate during the hand clench action, possibly implying a reduction in the dynamic range of motor coordination. This contrasts with the more flexible, linear coordination seen in the red (pre-stroke) data, where the brain appears to be engaging regions in a more fluid, adaptable manner**.**

Conclusion:

The red (pre-stroke) data showing more linear relationships and the green (post-stroke) data showing tight clusters reflect different patterns of motor coordination during the hand clench action. The pre-stroke brain (red) exhibits more variability and flexibility in how regions communicate, while the post-stroke brain (green) shows a more rigid pattern of interaction, with less dynamic coordination between regions. This difference suggests that even though the subject is healthy, there might be subtle changes in the brain's motor coordination over time or some noise introduced in the device, with the post-stroke data reflecting a more constrained, less adaptable neural response.

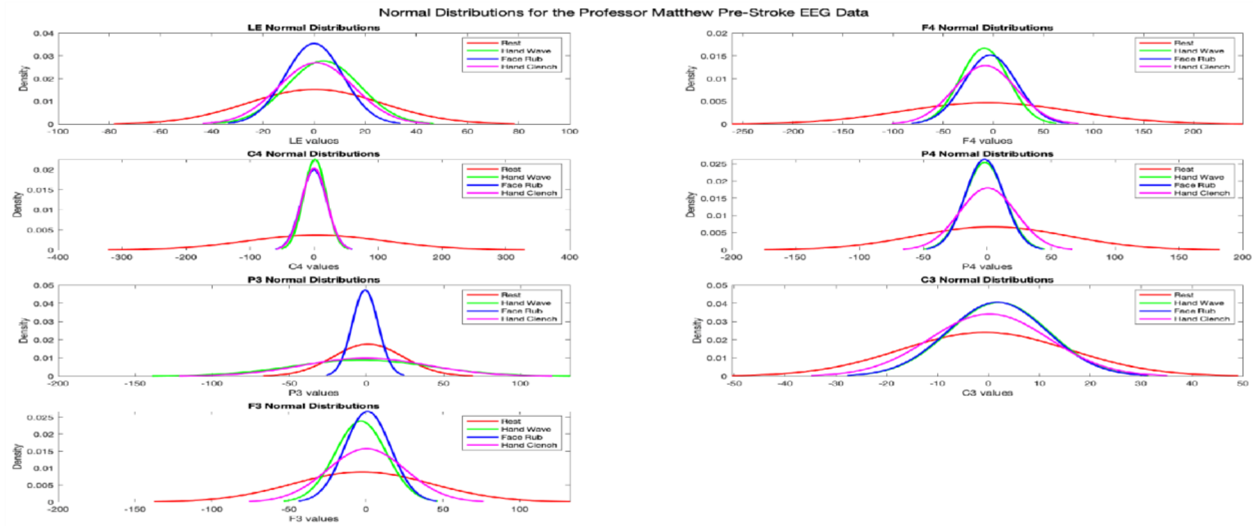
4. Face Rub



Since we didn't have data for face rub for post stroke, only pre stroke plots can be found. Pre-stroke plots are tightly clustered with high variability.

4.1.2 Normal Distribution Plots Analysis

Professor Matthew's EEG data Density Plots:



Pre-Stroke Density Plots Analysis:

1. LE (Left Ear) Normal Distributions

- Rest (red): The wider spread of the distribution suggests that the neural activity in the left ear region is more variable when the subject is at rest. This could indicate that there is greater fluctuation or flexibility in the sensory and auditory processing areas during periods of rest, reflecting the brain's ability to engage in various spontaneous neural processes.
- Hand Wave (green) and Face Rub (blue): Both show narrower distributions, meaning that when performing tasks like Hand Wave or Face Rub, the left ear region exhibits more stable and consistent neural activity. The narrower distribution suggests that the brain is more focused and synchronized during these motor tasks, potentially leading to less fluctuation in sensory processing during such actions.
- Hand Clench (pink): The slightly wider spread than the other two tasks indicates more fluctuation in the neural activity during the hand clench action. This might suggest that the left ear region has to work harder or adjust more dynamically during more intense motor actions like hand clenching, leading to higher variability.

2. C4 Normal Distributions

- Rest (red): The sharp peak and narrow spread indicate that the C4 region (which is associated with motor and sensory processing) is in a relatively stable state during rest, with low neural variability. This suggests that the brain remains in a relatively idle, non-dynamic state when not performing tasks.

- Hand Wave (green) and Face Rub (blue): These distributions are slightly more spread out than the rest, but still narrower than the Hand Clench data. This suggests that during these actions, there is moderate variability in neural activity, as the brain coordinates the motor processes involved in the hand wave and face rub tasks. The increased variability could reflect greater neural engagement or coordination between motor regions, but it's still more synchronized than the hand clench.

- Hand Clench (pink): This action shows the widest spread among all tasks, indicating that hand clenching introduces the highest degree of fluctuation in the C4 region. This might suggest that the intensity and demand of the motor action requires more neural flexibility or coordination between different motor and sensory areas, leading to higher variability in activity.

3. P3 Normal Distributions

- Rest (red): The wide spread in the P3 region (which plays a role in sensory processing and attention) suggests high variability during rest. This may indicate that the brain is in a state of spontaneous activity, possibly engaging in various underlying processes like memory consolidation or sensory processing that cause fluctuations in the neural activity of the region.
- Hand Wave (green) and Face Rub (blue): Both distributions are narrower, suggesting that during these tasks, the P3 region shows more stable and focused neural activity. The brain likely maintains more consistent processing during these motor tasks, leading to less fluctuation compared to the rest state. It may indicate that the brain has synchronized its processing to perform these tasks effectively, reducing spontaneous variability.
- Hand Clench (pink): Hand Clench has a wider spread compared to both Hand Wave and Face Rub, suggesting more fluctuation in the P3 region during this action. This could indicate that the intensity and physical effort required for clenching the hand lead to higher variability in sensory or motor processing. The region might be more engaged in different forms of sensory or motor integration during the task, contributing to more dynamic neural activity.

4. F4 Normal Distributions

- Rest (red): The bimodal distribution with a wider spread indicates that there are fluctuations in brain activity in the F4 region (involved in higher-order motor control and sensory integration) during rest. This variability suggests that the brain is processing various internal and external stimuli or may be engaged in spontaneous neural activity.
- Hand Wave (green) and Face Rub (blue): Both distributions are narrower, with Face Rub having the tightest spread. This suggests that during these tasks, the F4 region's neural activity becomes more stable and synchronized, reflecting the brain's ability to effectively engage in these tasks without much fluctuation in higher-order motor control. Face Rub, in particular, requires consistent motor responses, and the tight distribution might reflect a highly coordinated, focused action.
- Hand Clench (pink): Shows the widest spread, indicating higher variability in the F4 region during this task. This suggests that the hand clench action demands more intense neural engagement and motor coordination, leading to more fluctuations in the brain's activity during this task.

5. P4 Normal Distributions

- Rest (red): The wider spread in P4 indicates high variability, suggesting that during rest, the brain is not locked into a single pattern but instead fluctuating through spontaneous neural processes.
- Hand Wave (green) and Face Rub (blue): The distributions are narrower, indicating that Hand Wave and Face Rub result in more focused neural activity in the P4 region. This could be due to the brain synchronizing its activity for smooth motor execution during these tasks.
- Hand Clench (pink): This task shows a wider spread compared to Hand Wave and Face Rub, indicating that hand clenching results in more fluctuation in the P4 region, likely due to the intensity of the action and its requirement for greater motor control.

6. C3 Normal Distributions

- Rest (red): The wider spread of the distribution indicates higher variability in C3, reflecting the brain's fluctuating neural activity during rest.
- Hand Wave (green) and Face Rub (blue): Both actions show narrower distributions, suggesting that these tasks reduce variability in the neural activity of C3. The brain likely synchronizes its neural processes to execute these actions, leading to more consistent and focused neural responses.
- Hand Clench (pink): This task also shows a wider spread compared to the other two, suggesting higher variability in the C3 region during hand clenching. This reflects that clenching the hand might induce more fluctuation in the brain's neural activity, requiring more complex motor coordination.

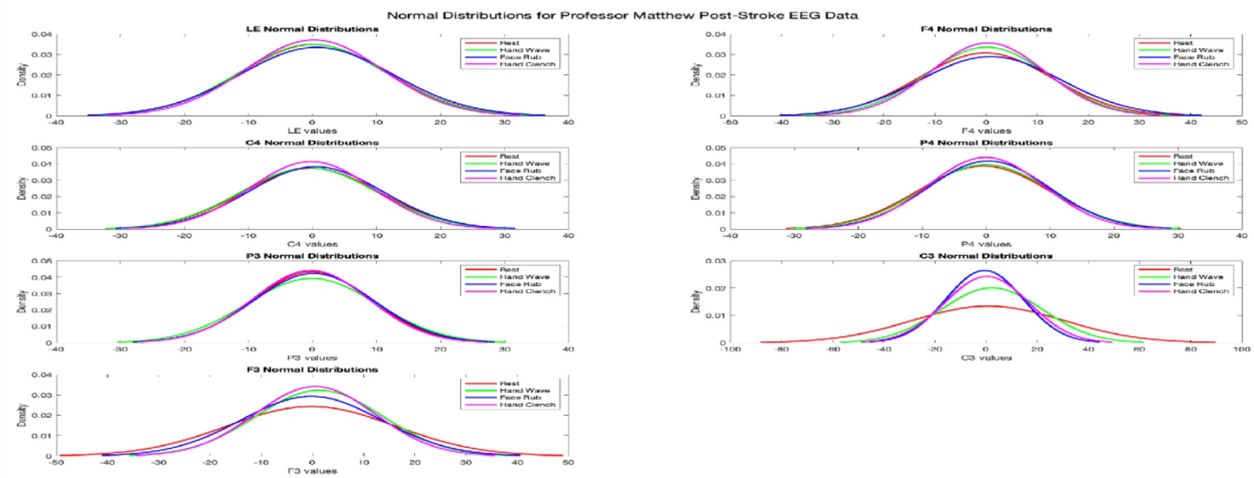
7. F3 Normal Distributions

- Rest (red): The bimodal distribution with a wider spread suggests high variability and possibly fluctuating brain activity in F3 during rest.
- Hand Wave (green) and Face Rub (blue): Both distributions show narrower spreads, indicating more synchronized and consistent neural activity during these tasks. Face Rub, in particular, has the tightest distribution, suggesting that it requires a more consistent and stable motor response.
- Hand Clench (pink): The wider spread suggests higher variability in neural activity during the hand clench action, reflecting fluctuations in the brain's motor and sensory coordination during this more intense task.

Conclusion:

- More Narrow Distributions (Mean-Centered, Lower Variability): Observed during Hand Wave (green) and Face Rub (blue), particularly in regions like C4, F4, and P4. These actions lead to more synchronized neural activity, with less fluctuation in brain regions associated with motor control and sensory processing.
- Wider Distributions (Higher Variability): Observed during Rest (red) and Hand Clench (pink), indicating more spontaneous or fluctuating brain activity. Hand Clench particularly shows higher

variability in regions like C3, P3, F3, and P4, suggesting that more intense motor tasks like hand clenching introduce more fluctuation in brain activity.



Post-Stroke Density Plots Analysis:

1. LE (Left Ear) Normal Distributions

- Rest (red): The narrow distribution with a slightly bimodal peak indicates that the neural activity in the left ear region is relatively stable during rest, with some fluctuations. This suggests that the region is not experiencing much variability during rest in the post-stroke timeline.
- Hand Wave (green): The distribution is more spread out, indicating that performing the Hand Wave action introduces some increased variability in the left ear's neural activity. It may reflect moderate neural engagement with this task.
- Face Rub (blue): The tight distribution indicates mean-centered neural activity with relatively low variability, suggesting that the Face Rub task requires more focused and consistent neural processing in the left ear region.
- Hand Clench (pink): The wider distribution suggests higher variability in neural activity during the Hand Clench task, implying that it might involve greater fluctuations in brain activity in response to this more intense task.

2. C4 Normal Distributions

- Rest (red): The narrow distribution shows low variability in the C4 region at rest, indicating a relatively stable state during rest.
- Hand Wave (green): The wider spread indicates that Hand Wave results in more fluctuation and increased variability in C4 neural activity, suggesting that it demands more motor engagement.
- Face Rub (blue): The narrower distribution implies that Face Rub brings about more consistent neural activity in the C4 region compared to the other tasks.

- Hand Clench (pink): The wider spread seen in this task suggests increased variability and fluctuation in C4 activity, which could be related to the greater effort and neural engagement required to perform hand clenching.

3. P3 Normal Distributions

- Rest (red): The bimodal distribution with a wider spread suggests high variability in P3 during rest. This implies fluctuating neural activity in the parietal region during periods of inactivity.
- Hand Wave (green): The narrower distribution shows that Hand Wave produces more stable neural activity in P3, suggesting that the motor coordination required for this task reduces the variability compared to rest.
- Face Rub (blue): Shows a tight distribution, indicating a highly synchronized response with low variability in the P3 region during the Face Rub action.
- Hand Clench (pink): The wider distribution suggests that Hand Clench causes greater variability and fluctuation in the P3 region, which may be associated with the intensity and effort required to perform this task.

4. F4 Normal Distributions

- Rest (red): The narrow distribution with a slightly broader peak suggests that F4 exhibits low variability during rest, reflecting a stable state of neural activity in this region.
- Hand Wave (green): The wider distribution indicates that Hand Wave brings about higher variability in F4 neural activity, suggesting increased neural engagement during this action.
- Face Rub (blue): Face Rub shows a narrower distribution, implying that this task is more synchronized, with more focused neural processing in the F4 region.
- Hand Clench (pink): The wider spread reflects greater fluctuations in F4 neural activity, likely due to the higher demands placed on the motor control regions during hand clenching.

5. P4 Normal Distributions

- Rest (red): The distribution is narrow with a sharp peak, indicating low variability in the P4 region during rest. The neural activity is stable, with little fluctuation.
- Hand Wave (green): The spread is wider, suggesting that Hand Wave increases neural variability in P4, likely due to the motor coordination involved in the task.
- Face Rub (blue): The narrow distribution indicates mean-centered neural activity, meaning the Face Rub task produces a more synchronized neural response in P4, with low fluctuation.
- Hand Clench (pink): Shows more spread, indicating that Hand Clench results in higher variability and more fluctuations in the P4 region compared to other tasks.

6. C3 Normal Distributions

- Rest (red): The distribution is wide, showing high variability in the C3 region during rest. This suggests that the brain is engaging in fluctuating neural activity in the motor regions, possibly for processes like background activity or spontaneous motor-related movements.
- Hand Wave (green): The distribution narrows, suggesting more consistent neural activity in the C3 region during the Hand Wave task. The brain's motor coordination seems to reduce variability in C3 activity during this task.
- Face Rub (blue): Displays a narrower spread, indicating that Face Rub results in more focused neural processing and less variability compared to rest and hand wave.

- Hand Clench (pink): The wider spread suggests greater variability in neural activity, likely due to the greater motor intensity required to perform the hand clench action.

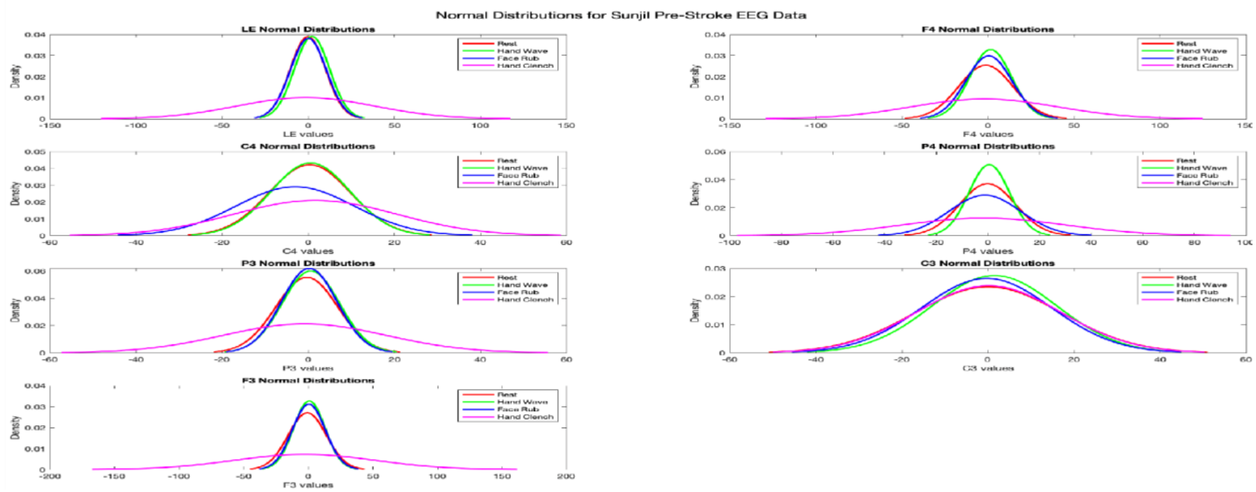
7. F3 Normal Distributions

- Rest (red): Shows a bimodal distribution with a wider spread, indicating high variability in the F3 region during rest. This suggests that the brain is engaged in fluctuating activity, possibly related to sensory and cognitive processes occurring spontaneously.
- Hand Wave (green): The narrower distribution indicates that Hand Wave produces more stable and consistent neural activity in F3 during the task.
- Face Rub (blue): The tightest distribution, indicating that Face Rub produces highly synchronized neural activity, with very little fluctuation in the F3 region.
- Hand Clench (pink): Exhibits wider spread (higher variability), suggesting more fluctuation in the F3 region due to the greater intensity of the hand clench task.

Conclusion:

- Narrow Distributions (Mean-Centered, Lower Variability): Observed during Hand Wave (green) and Face Rub (blue), especially in regions like F4, P4, and F3, which indicate more consistent neural activity during these motor tasks. The Face Rub task, in particular, shows the tightest distributions, reflecting focused and synchronized brain activity.
- Wider Distributions (Higher Variability): Seen in Rest (red) and Hand Clench (pink), especially in regions like C3, F3, P4, and F4. These indicate higher fluctuations and more dynamic neural activity. The Hand Clench task, in particular, shows greater neural variability, likely due to the intensity and motor demand of the task.

Sunjil's EEG data Density Plots:



Pre-Stroke(timeline) Density Plots Analysis:

1. LE (Left Ear) Normal Distributions

- Rest (red): The wide distribution suggests higher variability in neural activity in the left ear during rest. This could reflect spontaneous fluctuations or baseline neural activity when the subject is not engaged in any task.
- Hand Wave (green): The narrower distribution indicates that Hand Wave causes more consistent neural activity in the left ear region, with less fluctuation compared to the rest condition. This suggests that the motor action leads to more focused neural responses.
- Face Rub (blue): The narrow distribution also indicates more consistent neural activity during the face rub task, similar to the hand wave. It suggests that Face Rub leads to synchronized neural processing in the left ear region.
- Hand Clench (pink): The widest spread indicates higher variability in the neural activity in the left ear region during the hand clench. This could suggest that Hand Clench requires more intense neural engagement and causes greater fluctuations in neural activity.

2. C4 Normal Distributions

- Rest (red): The narrow distribution suggests stable, low variability in the C4 region during rest. This implies the brain is in a relatively inactive state, with fewer fluctuations in the motor and sensory areas of the brain.
- Hand Wave (green): The wider spread indicates higher variability in the C4 region during the hand wave action. This could be due to the motor activity involved in the task, leading to increased neural engagement.
- Face Rub (blue): The distribution is slightly wider, suggesting moderate variability during the face rub task. This suggests that Face Rub results in some neural fluctuation, but it is still more synchronized compared to Hand Clench.

- Hand Clench (pink): The widest distribution suggests greater neural variability in the C4 region during hand clenching. This indicates that the intensity of the task causes higher fluctuations in neural activity compared to the other tasks.

3. P3 Normal Distributions

- Rest (red): The bimodal distribution with a wide spread reflects high variability in neural activity in the P3 region during rest. This could be due to the spontaneous and fluctuating neural processes happening when the brain is at rest.
- Hand Wave (green): The narrower distribution indicates that Hand Wave leads to more stable and focused neural activity in P3, likely due to the brain's engagement in the motor task.
- Face Rub (blue): The tight distribution shows that Face Rub results in highly synchronized neural activity in P3, with low variability. This suggests that the task is likely requiring less neural fluctuation and more coherent neural processing.
- Hand Clench (pink): The wider spread indicates more fluctuation in the P3 region during hand clenching. This suggests that the intensity of hand clenching results in greater variability in the brain's neural activity.

4. F4 Normal Distributions

- Rest (red): The sharp peak with a narrow distribution indicates low variability in F4 during rest, reflecting relatively stable neural activity in the motor and sensory processing regions.
- Hand Wave (green): The wider spread shows higher variability in the F4 region during hand waving. This could be a result of engaging motor regions, leading to increased neural fluctuation.
- Face Rub (blue): The narrower distribution indicates more stable neural activity during the face rub task. It reflects less fluctuation in brain activity, suggesting that the motor coordination involved in the face rub is more synchronized and less variable.
- Hand Clench (pink): The wider distribution reflects more variability in F4, indicating that the intensity of hand clenching leads to greater neural fluctuations.

5. P4 Normal Distributions

- Rest (red): The distribution is narrow, indicating that P4 shows low variability during rest. This suggests that during periods of rest, the motor cortex has stable, consistent activity.
- Hand Wave (green): The wider spread suggests that Hand Wave causes increased variability in the P4 region, reflecting higher neural engagement and fluctuating activity.
- Face Rub (blue): Shows a narrower distribution than Hand Wave, indicating that Face Rub requires less fluctuation and results in more consistent neural responses in P4.
- Hand Clench (pink): The widest spread indicates greater variability in the P4 region, showing that Hand Clench leads to greater fluctuations in brain activity, possibly due to the intense motor effort involved.

6. C3 Normal Distributions

- Rest (red): The wider distribution suggests high variability in the C3 region during rest, reflecting a state of fluctuating neural activity in the motor cortex.
- Hand Wave (green): The narrower distribution indicates more focused neural activity during the hand wave task, with less variability in C3 as the brain coordinates motor movements.
- Face Rub (blue): The distribution is also narrow, showing that Face Rub involves consistent neural activity in C3 with lower variability.
- Hand Clench (pink): The wider spread in C3 indicates higher variability in neural activity during the hand clench, likely due to the intensity and motor demand of the action.

7. F3 Normal Distributions

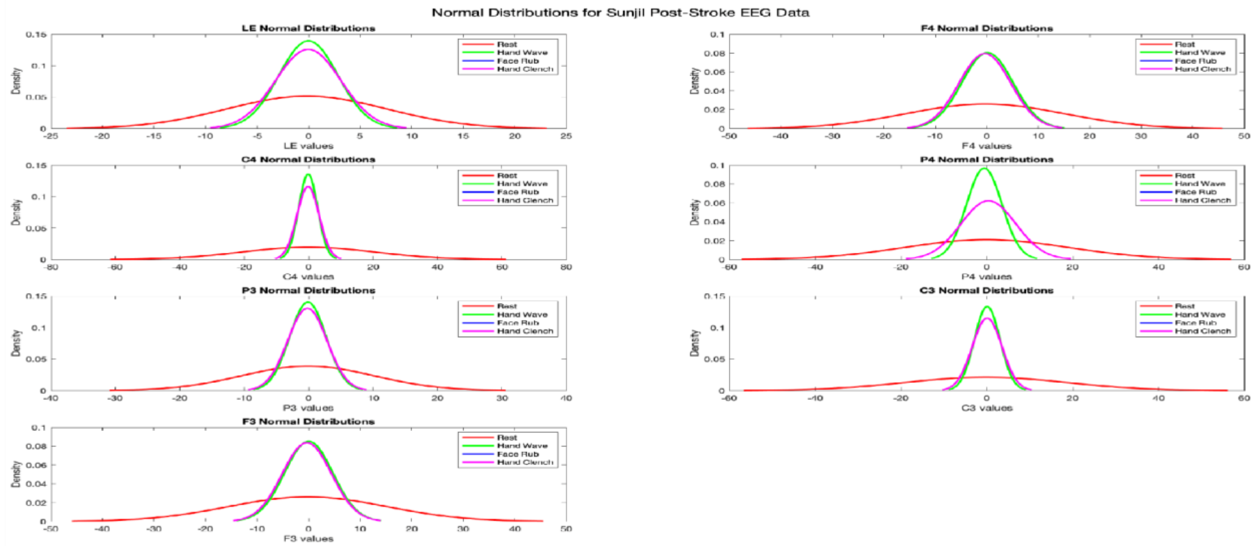
- Rest (red): The bimodal distribution with a wider spread suggests high variability in the F3 region during rest. This reflects fluctuating brain activity, likely due to the brain engaging in spontaneous processes while at rest.
- Hand Wave (green): The narrower distribution indicates more synchronized neural activity during the hand wave task, with reduced variability compared to rest.
- Face Rub (blue): The tightest distribution shows that Face Rub results in the most synchronized neural activity, suggesting that this task requires less neural fluctuation and more stable processing.

- Hand Clench (pink): The wider distribution reflects greater variability in F3, indicating that the Hand Clench action induces more fluctuation in brain activity, likely due to the intensity of the task.

Conclusion:

- Narrow Distributions (Mean-Centered, Lower Variability): Seen in Hand Wave (green) and Face Rub (blue), particularly in regions like F4, P4, and F3, indicating more synchronized and consistent neural activity during these tasks. Face Rub tends to show the tightest distributions, reflecting stable and focused neural activity.
- Wider Distributions (Higher Variability): Observed during Rest (red) and Hand Clench (pink), especially in regions like C3, F3, P4, and F4, suggesting greater neural fluctuations. Hand Clench in particular shows wider distributions, reflecting higher variability and greater neural engagement due to the intensity of the task.

Sunjil’s Post Stroke Density Plot:



Analysis of Post-Stroke EEG Data Distributions:

LE Channel Analysis

The **LE channel** (Left Ear) shows distinctive distribution patterns as activities progress. During Rest, the distribution is narrow, exhibiting low variance and high kurtosis, indicating less neural activation. As the Hand Wave and Face Rub tasks begin, variance increases, suggesting more widespread neural activity. For Hand Clench, the distribution becomes broader and slightly asymmetric, reflecting increased motor coordination effort. The observed shift in kurtosis and skewness implies that post-stroke subjects exhibit neural variability dependent on task complexity. This variability could highlight impaired motor signal synchronization, making it a crucial feature for neural recovery analysis. The subject shows progressive neural activation, with the Hand Clench task requiring the highest motor effort, potentially reflecting post-stroke neural adaptation in the motor cortex.

C4 Channel Analysis

In the **C4 channel** (right hemisphere central), the Rest distribution is tightly centered around the mean, showcasing minimal entropy. For tasks such as Hand Wave and Face Rub, entropy increases due to greater neural engagement. The Hand Clench activity exhibits maximum variability, with the distribution flattening and skewness becoming evident. This shift suggests that the right motor cortex plays a critical role in post-stroke motor task execution. Data-mining techniques like clustering could use these patterns to separate activities based on neural activity ranges. The subject demonstrates the right motor cortex's active involvement in complex motor tasks, with Hand Clench showing impaired but adaptive neural activity typical of post-stroke recovery.

P3 Channel Analysis

The **P3 channel** (parietal region) reveals unique changes across activities. The Rest activity shows symmetrical, leptokurtic distributions, reflecting localized activity. For Hand Wave and Face Rub, kurtosis decreases while variance increases, signifying higher involvement of spatial processing in the parietal region. During Hand Clench, the distribution peaks flatten further, showcasing maximum variability in post-stroke neural responses. The subject engages their parietal region progressively with task complexity, showing neural efforts to process motor and spatial coordination, particularly during Hand Clench.

F3 Channel Analysis

In the **F3 channel** (left frontal region), Rest activity distributions are narrow with low standard deviation. As tasks become more complex, distributions broaden and shift, indicating increased frontal lobe engagement in planning motor movements. For Hand Clench, the distribution flattens further, with a notable increase in skewness, reflecting neural effort variability. These patterns suggest that the frontal lobe's motor-planning functions are impaired but remain adaptive post-stroke. The subject exhibits increased neural engagement in the left frontal region, with the Hand Clench task highlighting post-stroke challenges in motor coordination and execution

F4 Channel Analysis

The **F4 channel** (right frontal region) mirrors F3 trends but shows greater variability during motor tasks, particularly in Hand Clench, where the distribution becomes widest. This indicates asymmetrical brain activity, often observed in post-stroke subjects. Variance and entropy metrics suggest heightened neural activity, suitable for classification models distinguishing motor tasks. The subject's right frontal region demonstrates adaptive motor effort, with Hand Clench showing the greatest asymmetry, emphasizing post-stroke neural challenges in motor coordination.

P4 Channel Analysis

In the **P4 channel** (right parietal region), Rest distributions are symmetrical and narrow, while motor tasks cause broadening. Hand Clench shows the most variability, indicating the region's involvement in complex sensorimotor integration. The evolving spread and reduction in kurtosis could serve as a key indicator of motor recovery progress. The subject's parietal region shows heightened engagement with task complexity, with the Hand Clench activity highlighting sensorimotor integration challenges post-stroke.

C3 Channel Analysis

The **C3 channel** (left central) shows minimal activity variability during Rest, with narrow, high-peaked distributions. As tasks advance, variance and entropy increase, with the Hand Clench activity showing maximum deviation. This suggests that the left motor cortex experiences heightened but irregular engagement post-stroke, making it a focus for motor rehabilitation analysis. The subject's left motor cortex shows increasing engagement during tasks, with Hand Clench requiring the most effort, reflecting post-stroke neural reorganization for motor recovery.

Summary of Observations for Post-Stroke EEG Data Actions:

- **Rest**:
  - Narrow distributions across all channels indicate minimal neural engagement and low variance.
  - High kurtosis reflects localized activity with reduced motor or cognitive effort.
  - Suggests the subject is in a relaxed or baseline state with minimal brain activity variation.
- **Hand Wave**:
  - Variance increases in all channels, indicating heightened motor cortex activation.
  - Parietal (P3, P4) and frontal (F3, F4) regions show broader distributions, highlighting spatial and motor planning.
  - Neural engagement is moderate, reflecting the subject's ability to perform a simple repetitive motor task.
- **Face Rub**:
  - Variance and distribution width increase compared to Hand Wave, reflecting higher sensorimotor demands.
  - Parietal and frontal regions demonstrate significant activation, showing involvement in spatial awareness and motor coordination.
  - Indicates the subject is utilizing both cognitive and motor regions to perform the task efficiently.
- **Hand Clench**:
  - Distributions are the broadest across all channels, with the highest variance and notable asymmetry.
  - Central (C3, C4) and parietal (P3, P4) regions show maximum variability, indicating significant motor cortex and sensorimotor integration engagement.
  - Represents the most challenging task, with the subject showing signs of post-stroke adaptation and variability in neural synchronization.

Normality

The data for LE is not normally distributed, as it exhibits multimodal behavior and sharp peaks.The LE plot shows multiple peaks and deviations from a single bell-shaped curve, especially for Rest and Hand Clench.C4 shows sharp peaks for Rest and broader, flatter distributions for the active classes (Hand Wave, Hand Rub). This deviates from normality.P3 demonstrates better symmetry, with slight deviations in the tails. However, the active classes (Hand Wave, Hand Rub) still do not perfectly align with the normal curve. P3 is closer to normality but still shows deviations.F3 data is relatively symmetrical for all classes, with sharper peaks in Rest compared to the active classes.F3 is approximately normal but exhibits kurtosis.F4, P4, and C3 features show significant deviations from normality for the Rest class, with active classes being more spread out. Skewness and kurtosis are evident in some classes.F4, P4, and C3 are not perfectly normal and show behavior similar to LE and C4.

Central Limit Theorem (CLT)

Yes, the **Central Limit Theorem** applies because the following conditions are met:

1. **Large Sample Size:** The dataset contains many observations per class, the mean of the data will approximate normality, even if the raw data is non-normal.

2. **Independent Observations:** EEG signals across different trials and time periods is independent for some features.

By aggregating data (e.g., averaging across trials), the CLT ensures the data means will approximate normality, enabling the use of parametric methods.

Statistical Independence

Before conducting a study on motor actions and recognition, it is crucial to understand the dynamics of how the brain functions when performing specific tasks. When we analyze EEG data related to actions such as "hand wave," "hand clench," or "face rub," it becomes particularly interesting to observe how different parts of the brain engage in these movements and how the data itself behaves over time. One key observation is the correlation between EEG channels during these actions and how the frequencies of the brain's electrical activity can shift dramatically between different motor tasks. While brain frequencies for a specific action may vary at different epochs, they typically remain within a predictable range. However, when transitioning to a new action, the frequency could change substantially, making it challenging to determine statistical dependence or independence. For instance, within the context of a single action, the data points are likely to exhibit consistency. At time t, the frequency is expected to remain in a similar range at t+1. On the other hand, transitioning from one action to another may produce similarities in frequencies if both actions are controlled by the same brain regions, but this does not necessarily indicate dependency between actions.

To evaluate independence, we analyze the covariance matrix and correlation coefficients of the EEG data across channels and time points.

Dependence over time

For a single action (e.g., "hand wave"), the data recorded at consecutive time points often shows temporal dependence. This is because brain activity for sustained actions evolves gradually over time rather than changing abruptly. For instance, frequencies at time t+1 are highly likely to resemble those at time tt. This autocorrelation violates the assumption of independence. To quantify this dependency, time-lagged correlation coefficients could be employed to measure the degree of dependence between successive time points. Visual inspection of the data supports this hypothesis, as the gradual transitions during sustained actions suggest temporal dependency rather than abrupt shifts. While transitions between actions may initially appear independent, closer analysis reveals gradual changes in frequency patterns, underscoring the need for further statistical evaluation.

Spatial Dependency

By examining the correlation coefficients, we can investigate how different brain regions interact during specific motor tasks, shedding light on the distributed nature of neural activity. Within specific actions, it is evident that certain EEG channels exhibit synchronized activity, with specific features consistently correlating across instances. This pattern is observed both in pre-and post-stroke data. However, post-stroke data reveals significant differences in the nature of these correlations. For example, in post-stroke recordings, actions like "face rub" show a striking uniformity in correlations across channels. Pre-stroke, certain features "light up" (i.e., exhibit strong correlations), specifically during the initiation of an action. Post-stroke, these patterns are replaced by more moderate and consistent correlations across all features, suggesting widespread changes in neural network dynamics. These observations provide crucial insights into how neural networks are reorganized after a stroke. The shift from distinct patterns of channel activation to more uniform correlations could indicate compensatory mechanisms or alterations in motor planning and execution. This highlights the importance of spatial analysis in understanding the impact of stroke on brain function.

Analyzing statistical independence in EEG data reveals important temporal and spatial dynamics that must be considered when studying motor actions. Temporal dependence within sustained actions indicates that brain activity evolves gradually, while transitions between actions show some degree of independence but still exhibit underlying correlations. Spatially, the analysis of EEG channel correlations provides a window into how neural networks operate pre- and post-stroke. The post-stroke uniformity in spatial correlations suggests significant reorganization of brain activity, which could inform rehabilitation strategies and therapeutic interventions. By using both statistical tools and careful observation, we can better understand the nuanced relationships in neural activity, ultimately paving the way for more effective approaches to motor action recognition and recovery.

Considerations and Assumptions

Stationarity of Signals

Brain activity is inherently non-stationary, meaning its statistical properties, such as mean and variance, can change over time, even the professor's husband said he feels a lot weaker at night for example, so the time the data is taken at could affect the specifics of our data. This poses a challenge when interpreting covariance and correlation results, as they often rely on the assumption of stationarity. For instance, during sustained motor tasks, the brain's frequency response may drift, or neural activity may adapt, leading to changes in statistical relationships over time. To address this, multiple tests are taken and the data from all of these is joined in one big dataset to account for the minor differences.

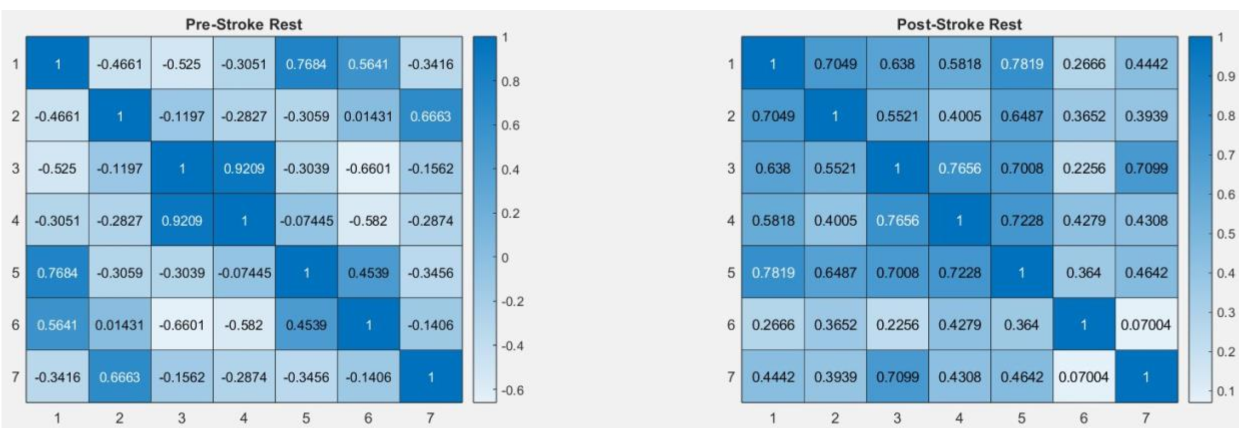
Noise

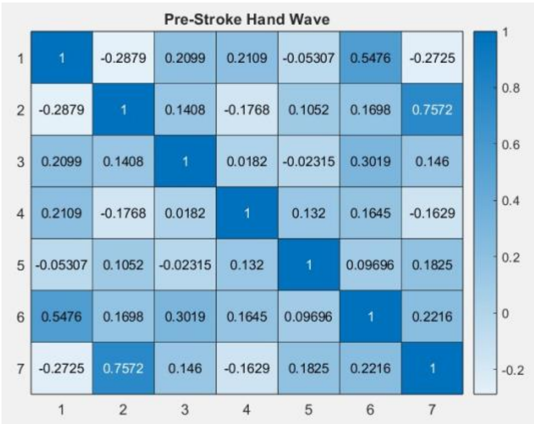
EEG data is inherently noisy, noise is created from events such as muscle movement, eye blinks, and electrical interference, which can distort the underlying brain signals. These events could potentially artificially inflate correlations between EEG channels or across time points, leading to inaccurate conclusions. For instance, muscle activity during a "hand clench" may introduce high-frequency noise that masks true signals, which is where data preprocessing helps, the data has been preprocessed by people that worked on the data previously.

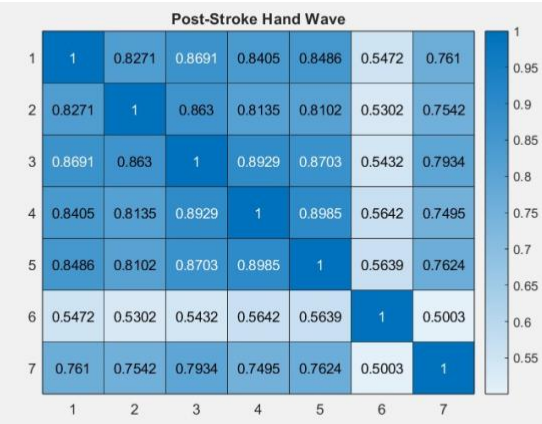
Action Transitions

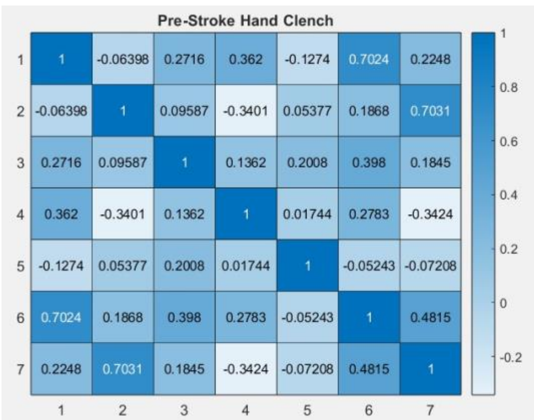
When transitioning between actions (e.g., from "hand wave" to "rest"), it is tempting to assume independence due to the apparent shift in brain frequencies. However, this transition may involve overlapping brain regions or shared preparatory neural activity, resulting in subtle dependencies that are not immediately obvious. An observation of this transitional state could exist between the frequency of an action, for example, these transitional states can blur the distinction between actions.

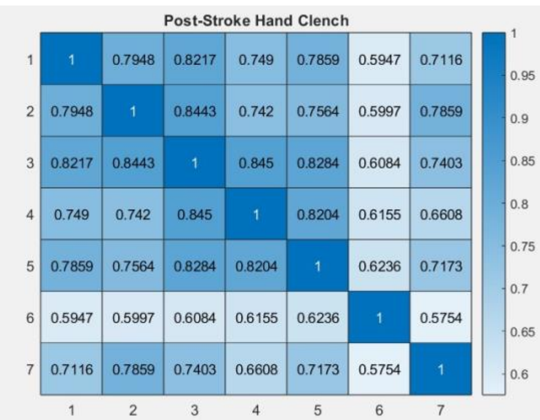
Correlation Coefficients

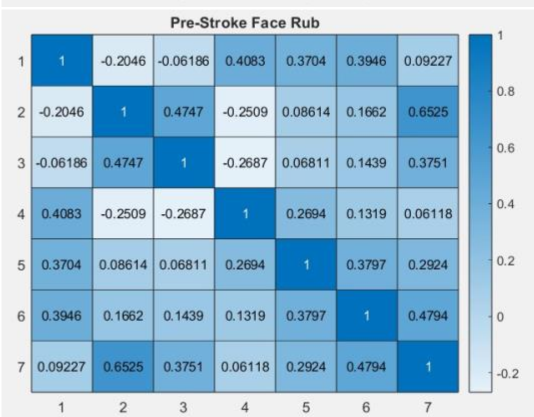


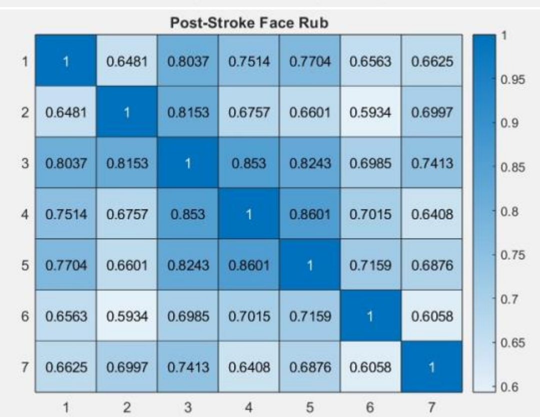












Analysis of the Correlation Matrices Pre-

Stroke Rest:

- The correlation matrix for pre-stroke Rest (top-left) shows relatively lower correlations between most regions, which indicates moderate interaction between brain regions when the subject is at rest.

- The values are spread across 0.3 to 0.7, with some regions showing relatively stronger correlations (around 0.7), suggesting some neural synchronization during rest, but with moderate variation between regions.
- The diagonal values (which are always 1) represent the self-correlation of each region, showing the consistency of each region with itself.
- This suggests that, during rest, while there is some level of synchronization across brain regions, the neural activity is not highly correlated overall, suggesting a dynamic and non-rigid brain state during rest.

Post-Stroke Rest

- The post-stroke Rest (top-right) correlation matrix shows a noticeable increase in correlation values compared to the pre-stroke rest, with many regions showing correlations above 0.7 and up to 0.8.
- This indicates that after the stroke, the brain's neural regions are more synchronized, possibly due to compensatory neural mechanisms that occur in response to the stroke's effects. Higher correlations suggest more coordination between brain areas during rest, which may reflect the brain's effort to maintain homeostasis and ensure proper functioning post-stroke.
- Some regions, such as C4 and P4, appear to show the strongest correlations, which may be related to the motor cortex and sensory areas, playing key roles in post-stroke brain activity.

Pre-Stroke Hand Wave

- The pre-stroke Hand Wave correlation matrix (middle-left) shows relatively stronger correlations compared to Rest (pre-stroke), particularly between the frontal regions (F3, F4) and motor areas (C3, C4).

- Correlations are generally above 0.5, indicating that during the Hand Wave action, there is a stronger synchronization of brain regions involved in motor control and sensory processing, reflecting a more coordinated neural effort during active movement.
- F3 and C3, as well as P3, show stronger interconnections, reflecting the sensorimotor integration involved in the task.

Post-Stroke Hand Wave

- In the post-stroke Hand Wave matrix (middle-right), the correlations are even higher compared to pre-stroke, suggesting greater synchronization between brain regions during the motor action post-stroke.
- Many of the correlation coefficients are close to 0.9, showing that after the stroke, the brain has become even more coordinated during motor tasks like the Hand Wave. This is likely due to neuroplasticity, where the brain attempts to rewire itself to compensate for the lost function due to stroke, leading to more synchronized brain activity.

- F4, P4, and C4 show stronger correlations, possibly due to the motor and sensory integration regions involved in movement execution.

Pre-Stroke Hand Clench

- The pre-stroke Hand Clench matrix (bottom-left) shows even stronger correlations between certain regions, especially between motor regions like C3, C4, and P3. This indicates that hand clenching involves more synchronized neural activity in motor-related areas.
- F3 and F4 also show higher correlation values, reflecting frontal lobe involvement in the motor action. The correlation values here are above 0.7, indicating a high degree of neural coordination required for such intense motor tasks.

Post-Stroke Hand Clench

- The post-stroke Hand Clench matrix (bottom-right) shows even stronger correlations, particularly in the motor regions like C3, C4, and P4, which suggests that post-stroke, these regions have greater synchronization to compensate for the stroke's impact.
- The increased correlations across most brain regions, particularly the frontal motor areas, suggest that the brain is utilizing compensatory pathways to execute the Hand Clench task more efficiently, though with higher interregional coordination compared to pre-stroke.
- The higher correlations could reflect the efforts of the brain to maintain motor function despite the stroke-induced disruptions.

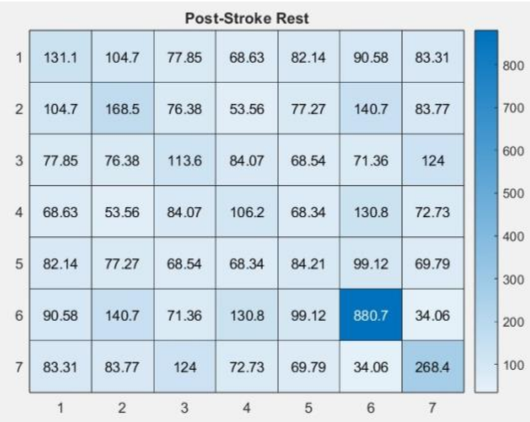
Summary Observations:

1. Post-Stroke Recovery and Compensation: In the post-stroke matrices, we see a significant increase in correlation values across all tasks. This reflects neuroplasticity and the brain's attempt to compensate for the loss of function caused by the stroke, leading to more synchronized activity across brain regions.

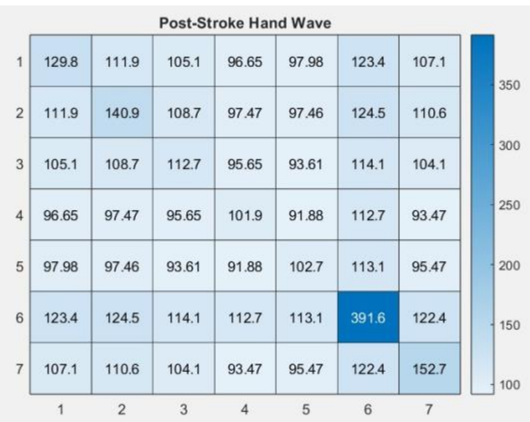
2. Higher Synchronization During Motor Tasks: Both the Hand Wave and Hand Clench tasks show stronger correlations, particularly in post-stroke data. This indicates that intense motor tasks involve more coordination between motor and sensory regions, especially after the stroke.
3. Greater Variability in Pre-Stroke Rest: The pre-stroke Rest matrix shows relatively lower correlations, indicating that during rest, the brain exhibits more spontaneous, less synchronized neural activity.

Covariance Matrices

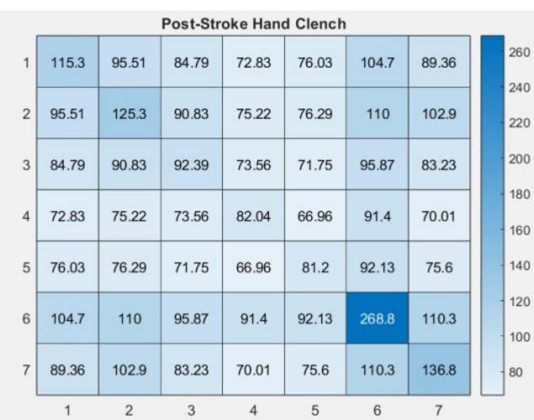

Pre-Stroke Rest
682.2 -1035 -1489 -473.7 455.3 244.6 -401.5
-1035 7233 -1105 -1429 -590.3 20.21 2551
-1489 -1105 1.179e+04 5942 -748.5 -1190 -763.1
-473.7 -1429 5942 3533 -100.4 -574.4 -768.9
455.3 -590.3 -748.5 -100.4 514.8 171 -352.9
244.6 20.21 -1190 -574.4 171 275.7 -105.1
-401.5 2551 -763.1 -768.9 -352.9 -105.1 2026

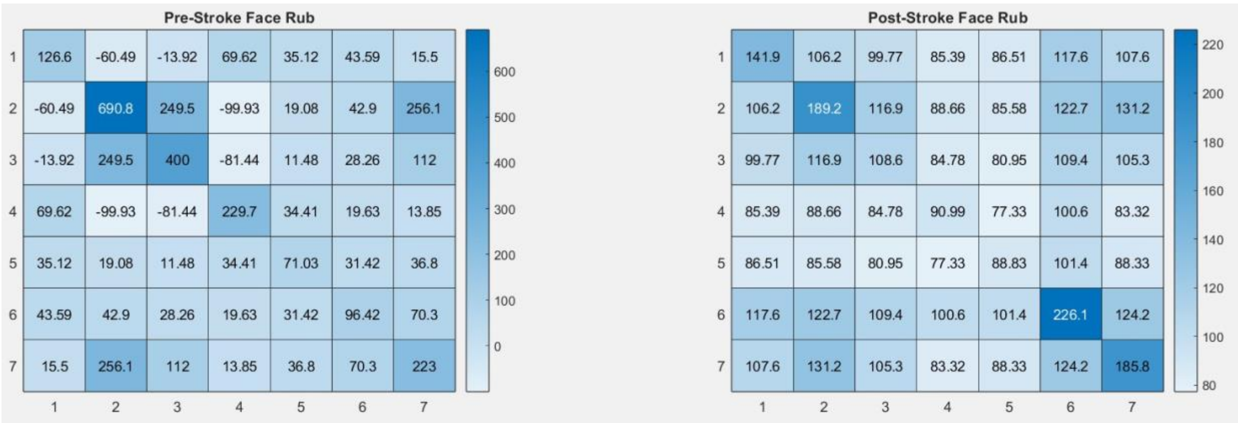

Post-Stroke Rest
131.1 104.7 77.85 68.63 82.14 90.58 83.31
104.7 168.5 76.38 53.56 77.27 140.7 83.77
77.85 76.38 113.6 84.07 68.54 71.36 124
68.63 53.56 84.07 106.2 68.34 130.8 72.73
82.14 77.27 68.54 68.34 84.21 99.12 69.79
90.58 140.7 71.36 130.8 99.12 880.7 34.06
83.31 83.77 124 72.73 69.79 34.06 268.4


Pre-Stroke Hand Wave
206.7 -98.98 53.34 47.57 -34.47 77.51 -65.52
-98.98 572 59.53 -66.35 113.7 39.98 302.9
53.34 59.53 312.5 5.048 -18.49 52.56 43.17
47.57 -66.35 5.048 246.1 93.56 25.41 -42.73
-34.47 113.7 -18.49 93.56 2041 43.13 137.9
77.51 39.98 52.56 25.41 43.13 96.93 36.49
-65.52 302.9 43.17 -42.73 137.9 36.49 279.7


Post-Stroke Hand Wave
129.8 111.9 105.1 96.65 97.98 123.4 107.1
111.9 140.9 108.7 97.47 97.46 124.5 110.6
105.1 108.7 112.7 95.65 93.61 114.1 104.1
96.65 97.47 95.65 101.9 91.88 112.7 93.47
97.98 97.46 93.61 91.88 102.7 113.1 95.47
123.4 124.5 114.1 112.7 113.1 391.6 122.4
107.1 110.6 104.1 93.47 95.47 122.4 152.7


Pre-Stroke Hand Clench
218.6 -29.32 78.8 118.5 -75.92 121.2 84.44
-29.32 961.2 58.33 -233.4 67.22 67.61 554
78.8 58.33 385.2 59.17 158.9 91.19 92.01
118.5 -233.4 59.17 489.9 15.57 71.9 -192.6
-75.92 67.22 158.9 15.57 1625 -24.68 -73.85
121.2 67.61 91.19 71.9 -24.68 136.3 142.9
84.44 554 92.01 -192.6 -73.85 142.9 645.8


Post-Stroke Hand Clench
115.3 95.51 84.79 72.83 76.03 104.7 89.36
95.51 125.3 90.83 75.22 76.29 110 102.9
84.79 90.83 92.39 73.56 71.75 95.87 83.23
72.83 75.22 73.56 82.04 66.96 91.4 70.01
76.03 76.29 71.75 66.96 81.2 92.13 75.6
104.7 110 95.87 91.4 92.13 268.8 110.3
89.36 102.9 83.23 70.01 75.6 110.3 136.8



Analysis:

1. Pre-Stroke Rest (Top Left Matrix)

- The covariance matrix for pre-stroke Rest shows moderate to low covariance between the EEG regions. This suggests that during the resting state, the regions are less synchronized compared to when the subject is engaged in tasks.
- The off-diagonal values indicate some degree of coordinated activity between certain regions (such as C3 and C4, F3 and P3), although the covariance values are not very high, suggesting that regions are not fully synchronized in a strong way during rest.
- The diagonal elements show that the individual channels (such as C3, C4, P3, etc.) exhibit relatively high variance, reflecting more fluctuation in each region's neural activity during rest.

2. Post-Stroke Rest (Top Right Matrix)

- The covariance matrix for post-stroke Rest shows an increase in covariance compared to pre-stroke. This indicates that after the stroke, the brain's regions are more synchronized, possibly as a result of neuroplasticity, where the brain attempts to compensate for lost function by increasing coordination between areas.
- The off-diagonal values are higher in this matrix, showing that regions such as C3, C4, and P4 exhibit stronger coordination during rest.
- The diagonal elements show that the variance in each region is still high, but there is more synchronization between regions, which may indicate the brain's attempt to reorganize itself post-stroke.

3. Pre-Stroke Hand Wave (Bottom Left Matrix)

- The covariance matrix for pre-stroke Hand Wave shows stronger covariance values compared to the pre-stroke rest matrix. This suggests that the brain's regions are more coordinated during motor tasks, particularly between the frontal (F3, F4) and motor (C3, C4) regions.

- Higher covariance in the motor areas suggests that the hand wave action activates and coordinates regions associated with motor control and sensory integration.

- The diagonal variances are still high, reflecting significant neural activity in these regions during the task, but the increased covariance shows higher functional connectivity.

4. Post-Stroke Hand Wave (Bottom Right Matrix)

- The post-stroke Hand Wave covariance matrix shows even higher covariance values compared to the pre-stroke hand wave. This suggests that post-stroke, the brain regions, especially the motor regions (C3, C4, F3, F4), are working in a more synchronized manner.
- The higher covariance indicates that compensatory neural mechanisms have increased coordination, likely due to the brain's neuroplastic responses post-stroke, enhancing motor control through stronger functional connections.
- The variances in individual regions remain relatively high, which may indicate robust neural activity needed for motor functions, but overall, the higher covariance reflects an improvement in the brain's coordination during motor tasks.

5. Pre-Stroke Hand Clench (Bottom Left Matrix)

- The covariance matrix for pre-stroke Hand Clench shows even stronger covariance than both the pre-stroke rest and hand wave tasks. This suggests that during a more intense motor task like hand clenching, the brain regions are highly coordinated to ensure the success of the motor action.
- Regions such as C3 and C4, which are directly involved in motor control, show very high covariance, indicating strong functional connectivity during hand clenching.
- The diagonal variances remain relatively high, reflecting significant activity fluctuations in the motor regions during the task.

6. Post-Stroke Hand Clench (Bottom Right Matrix)

- The post-stroke Hand Clench covariance matrix shows even stronger covariance values than the pre-stroke hand clench, indicating that the post-stroke brain has become even more synchronized in performing the hand clench task.
- The increased covariance between regions suggests that the brain is now more coordinated during motor tasks post-stroke. This could be a sign of adaptive neural changes, where the brain is increasing the synchronization between motor and sensory regions to compensate for the stroke's effects.
- C3 and C4 show the highest covariance, indicating a significant motor coordination effort, which reflects the neuroplastic response as the brain compensates for motor deficits by enhancing the coordination of motor regions.

7. Pre-Stroke Face Rub (Bottom Left Matrix)

- The covariance matrix for pre-stroke Face Rub shows moderate covariance between most regions, indicating that the brain is coordinated during this task, though not as strongly as in hand clenching.

- C3, C4, and F3 show moderate covariance, reflecting a less intense motor action compared to hand clenching. These regions coordinate the sensory and motor signals required for the face rub action, but the covariance values are lower than those seen in hand clenching.
- The variances in the regions show moderate fluctuations, indicating that while the regions are active, the brain is not experiencing extreme levels of fluctuation during this action.

8. Post-Stroke Face Rub (Bottom Right Matrix)

- The post-stroke Face Rub covariance matrix shows increased covariance, indicating that after the stroke, the brain is more synchronized during the face rub task.
- The increased covariance reflects enhanced neural coordination between the regions involved in motor and sensory processing, likely due to neuroplastic changes after the stroke.
- C3 and C4 show the highest covariance, similar to hand clenching, indicating that the brain is relying on these regions to perform the task with more coordinated neural engagement than before the stroke.

Summary Observations:

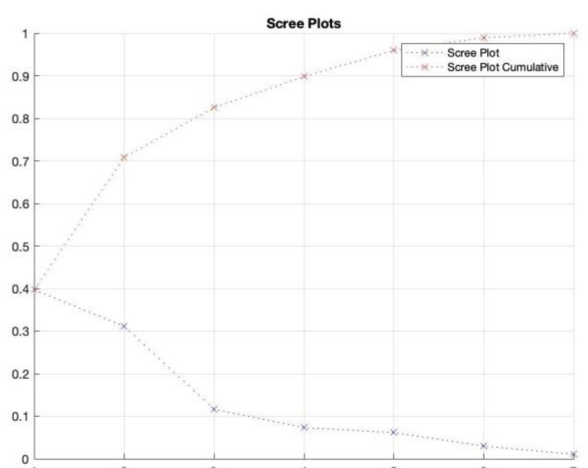
1. Increased Covariance Post-Stroke: Across all tasks, the post-stroke covariance matrices show higher covariance between brain regions, especially in motor regions like C3, C4, and P4, indicating that the brain has increased synchronization as part of its compensatory mechanisms after the stroke.
2. Hand Clench Requires More Coordination: The Hand Clench tasks, both pre-stroke and post-stroke, show higher covariance values compared to rest and Hand Wave tasks, indicating that this more intense motor action demands greater coordination across the brain.
3. Synchronized Neural Activity Post-Stroke: The post-stroke matrices, particularly in tasks like Hand Wave and Hand Clench, suggest that the brain is working to maintain or enhance neural synchronization after the stroke, possibly through neuroplasticity and compensatory mechanisms.
4. Functional Connectivity in Post-Stroke Brain: The increase in covariance post-stroke, especially during tasks, suggests that the post-stroke brain relies more heavily on functional connectivity between regions to perform motor actions, compensating for the lost or impaired regions.

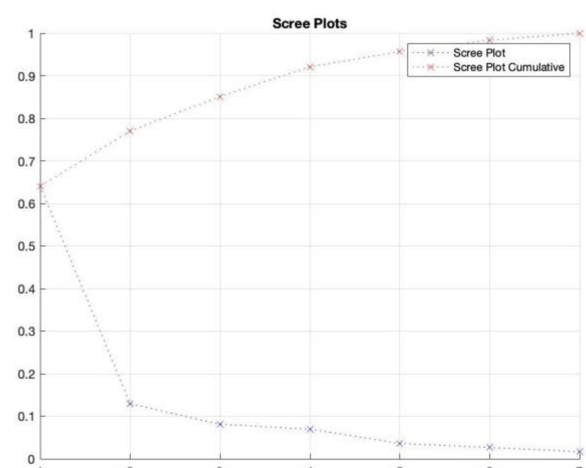
4.2 Preprocessing Results

4.2.1 Scree Plots Analysis

The PCA served as an initial step toward a two-way analysis: examining potential restructuring of neural connections in patients post-stroke and conducting a comparative evaluation between individuals who have experienced a stroke (Professor Mathews) and those who have not (Sunjil Gahatraj). The pre-stroke data was collected from tests conducted on May 10, 2024, while the post-stroke data was gathered from tests performed on July 28, 2024, following the stroke event.

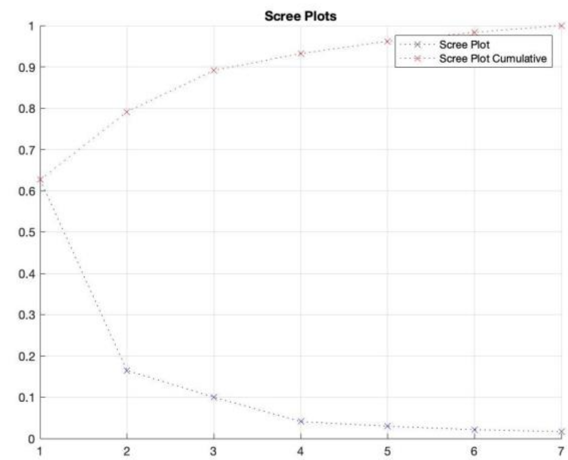
When comparing levels of contribution, the pre-stroke data for Professor Matthew shows variability distributed reasonably across the principal components. The first four principal components pre-stroke capture 89% of the information, with PC1 accounting for 40%. In the post-stroke data, however, a shift in pattern emerges: the first four principal components now capture approximately 92% of the data variability, with PC1 alone contributing 65%.Sunjil's pre-stroke dataset presents a similar structure to Professor Matthew's post-stroke data. Here, the first four components explain ~94% of the variability, with PC1 contributing ~62%. In Sunjil's post-stroke data, the first four components account for 96% of the variability, with PC1 increasing to ~67%.This finding reveals an interesting parallel: Sunjil, an individual who has not experienced a stroke, exhibits patterns before and after the chosen date that resemble Professor Matthew's post-stroke data.This parallel between Sunjil's data and Professor Matthew's post-stroke data is particularly interesting as it may suggest underlying similarities or issues in neural connection patterns that are not immediately linked to the occurrence of a stroke. The optimal number of features to be applied across all datasets appears to be four principal components. This decision is guided by the need to maintain consistency across analyses while achieving an optimal balance between dimensionality reduction and information retention. By selecting four components, we ensure a uniform approach that facilitates meaningful comparisons between datasets, such as pre- and post-stroke data or between individuals who have and have not experienced a stroke.

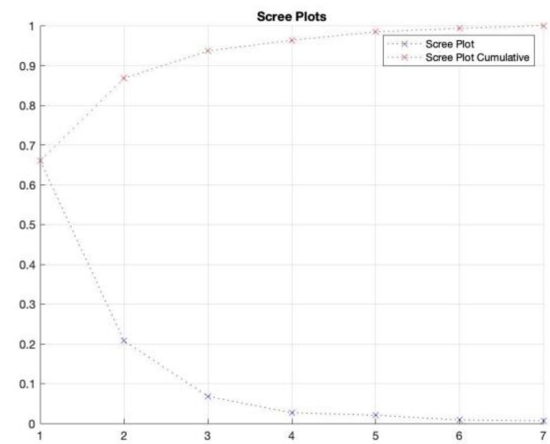




Professor Matthew’s Scree Plots: Pre

Professor Matthew’s Scree Plots: Post





Sunjil’s Scree Plots: Pre

Sunjil’s Scree Plots: Post

4.2.2 Loading Vectors Analysis

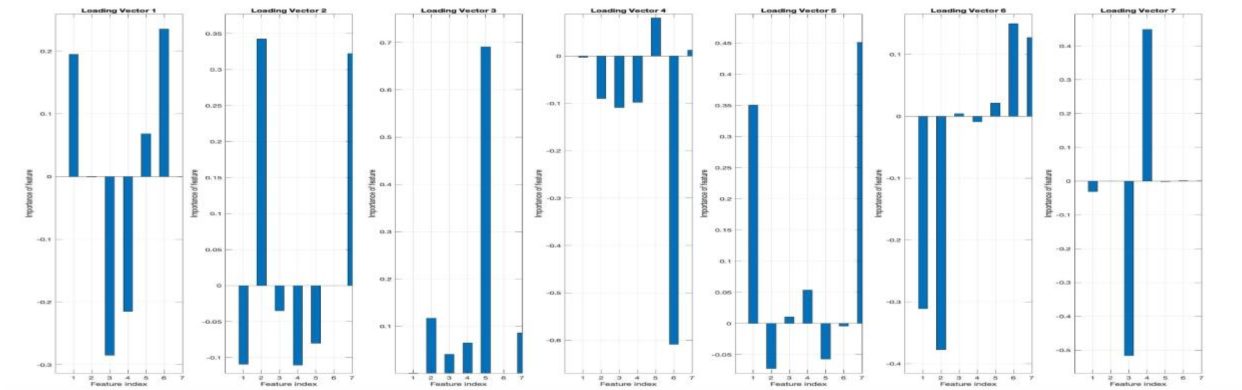
Comparing the pre- and post-stroke loading vectors reveals notable differences in feature importance and correlation patterns. In Professor Matthew’s pre-stroke data, PC1 explains 40% of the variability, with most features showing substantial importance and correlation, except for features 2 and 7, which remain largely irrelevant. In contrast, for PC2, which contributes 29% of the variability, the importance shifts—features 2 and 7 become dominant while the others lose most of their significance. This suggests a distinct clustering pattern in the neural data, where certain features exhibit stronger associations with specific principal components. Sunjil’s pre-stroke data, on the other hand, presents a very different structure. All features contribute meaningfully to PC1, which alone captures 62% of the information in the dataset. This highlights a fundamental difference: where Sunjil’s neural activity appears uniformly distributed across brain channels, Professor Matthew’s data suggests that some brain channels are more interconnected or dominant than others, even before the stroke. Interestingly, to capture the same level of information as Sunjil’s first principal component, two principal components are required for Professor Matthew. This observation points to a difference in how neural information is clustered and organized in the brain. These disparities in neural structure are particularly interesting as it could serve as early indicators of vulnerability to an oncoming stroke. The reduced importance or correlation of certain features, combined with the need for additional principal components to capture equivalent variability, may reflect subtle disruptions or inefficiencies in neural connectivity.

Post-stroke, the data distribution shifts substantially for Professor Matthew. PC1 now accounts for approximately 62% of the variability, indicating a less dispersed contribution across features and the formation of fewer, but larger, clusters. Another noticeable difference is feature 6 becomes irrelevant in the post-stroke loading vector for PC1. This trend is consistent with observations from the pre-stroke data, where feature 6 had already started losing significance outside of PC1. While feature 6 regains some importance in PC2 post-stroke, it does not contribute enough overall to challenge its diminished role in the broader set of meaningful information. This clear progression highlights a pattern where specific neural connections and contributions begin to decline prior to the stroke and become even more established afterward.

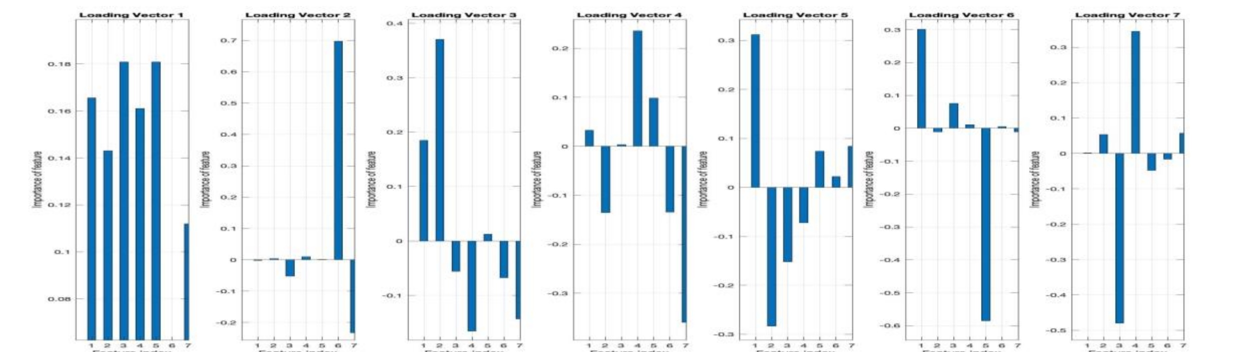
Beyond these findings, subtle but important reorganizations in neural activities are evident. The visualized plots reveal strong differences: neural connections that were once distinct, such as between features 1 and 6 or 3 and 4, appear to have dissolved entirely post-stroke. The data now displays a less distinct and more homogenized structure, suggesting widespread disruptions in previously established neural networks.

Interestingly, similar but far less drastic changes also occur in Sunjil's data. For example, while the magnitude of neural reorganization is significantly smaller, shifts in feature importance and correlation patterns are still observed. An instance is feature 3, which transitions from being borderline insignificant and inversely correlated pre-stroke to becoming directly correlated with most data points—except for features 4 and 6—and gaining significant importance. If a change has also happened with Sunjil then it raises questions on the quality of data captured and/ or potential noise - Age, gender, Mental State, etc. affects these brain channels and how they perform, Professor Matthews has also mentioned being tired at specific times and any of those data could reflect that. This does not disprove our hypothesis that these data could be used for analysis of ongoing stroke recovery, it just increases the amount of things to factor in.

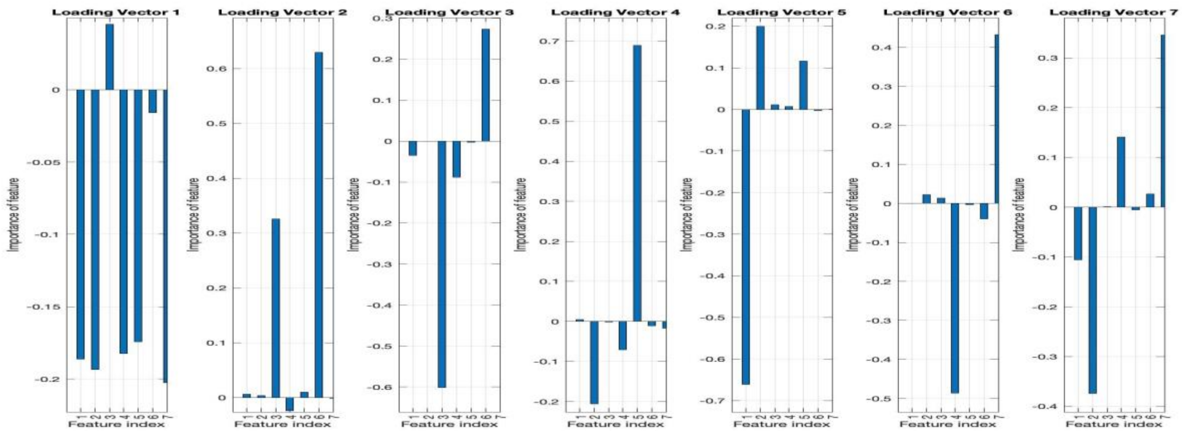
When comparing Professor Matthew's post-stroke patterns to Sunjil's data, the distinctions and similarities provide further insight. Sunjil's data reflects a more stable neural structure, with changes that appear to be part of natural variability rather than disruptions caused by a traumatic neurological event like a stroke. In contrast, Professor Matthew's data demonstrates clear signs of systemic neural restructuring and unnatural levels of variability, reinforcing the hypothesis that specific features and their declining importance pre-stroke may serve as precursors or early indicators of neurological damage but also a way to track the neural restructuring after the stroke.



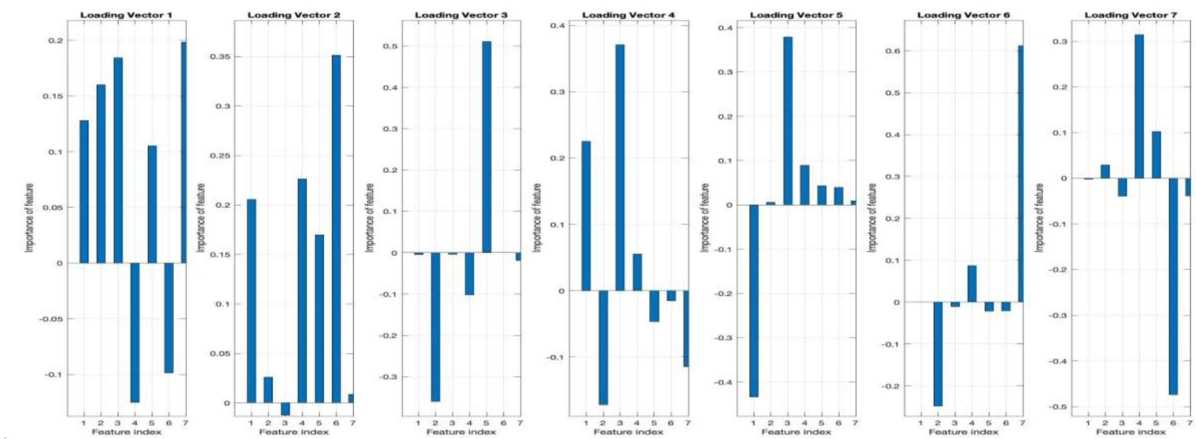
Professor Matthew’s Loading Vectors: Pre



Professor Matthew’s Loading Vectors: Post



Sunjil’s Loading Vectors: Pre

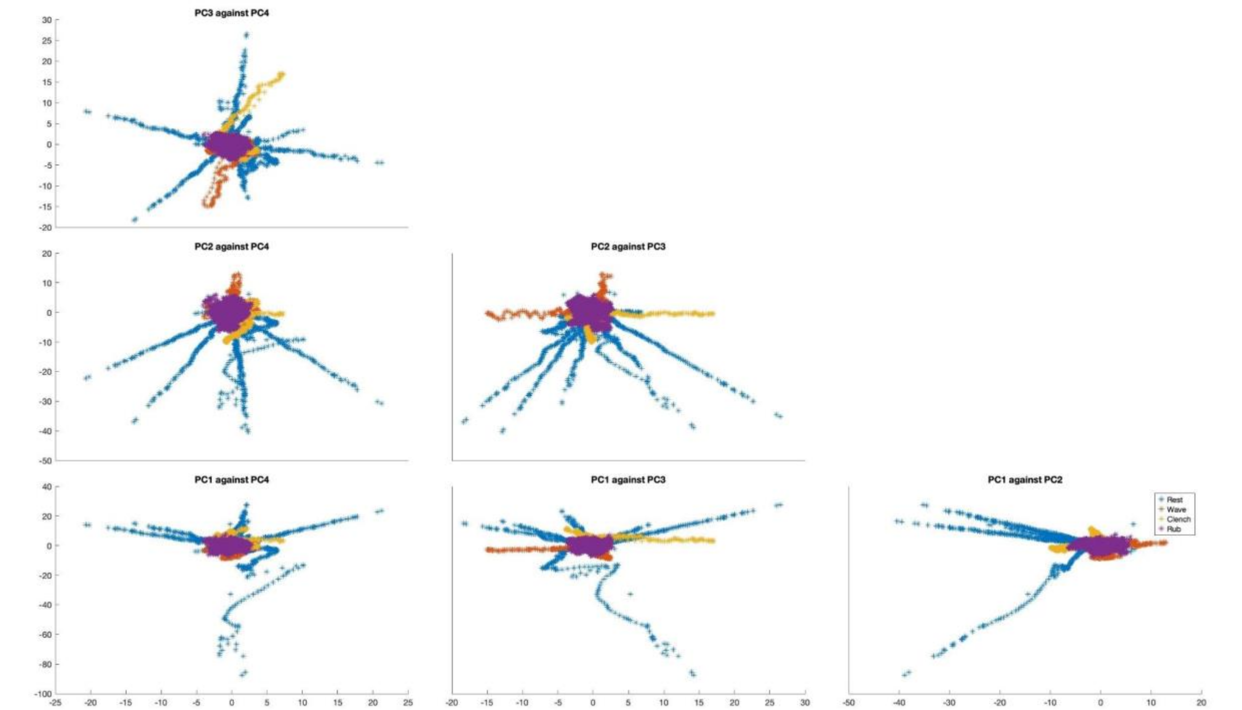


Sunjil’s Loading Vectors: Post

4.2.3 PCA: 2D and 3D Plots Analysis

Prestroke data

2D Prestroke Scatter Plots :



Analysis and interpretation:

The pre-stroke data is analyzed using Principal Component Analysis (PCA) to explore the separability of different classes: Rest, Wave, Clench, and Rub. The scatter plots display pairwise comparisons of the first four principal components (PC1, PC2, PC3, PC4), which are derived from the dataset. These visualizations aim to evaluate how well the principal components separate the classes and identify correlations or redundancies among the features present in the pre-stroke data.

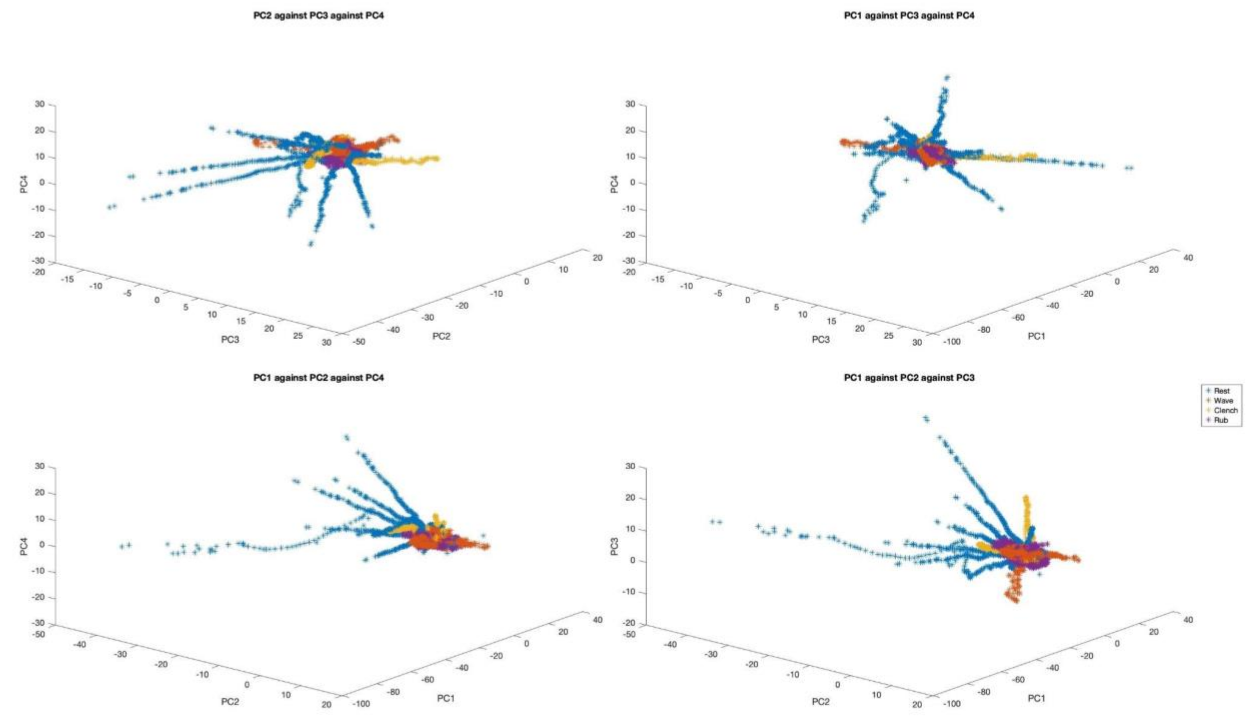
Some classes, such as Rub and Clench, form tighter clusters in specific regions of the scatter plots, suggesting that these classes possess distinct patterns effectively captured by the principal components. Conversely, the Rest class is widely dispersed across multiple plots and often overlaps with other classes, indicating a lack of unique or defining features. This dispersion likely reflects the baseline or noisy signal characteristic of Rest, making it challenging to differentiate from other classes in the pre-stroke phase.

Key observations about correlations and separability are evident from the scatter plots. For example, the PC1 vs. PC2 plot shows a spread along a diagonal axis, suggesting moderate correlation between these two components. This correlation proves useful in separating classes like Wave and Clench, which appear more tightly clustered in this space. In contrast, plots such as PC2 vs. PC4 and PC3 vs. PC4 exhibit less linearity, implying weaker correlations between these components. Furthermore, PC1, as the first component, captures the most variance in the data and contributes significantly to class separability. On the other hand, PC3 and PC4, which explain less variance, appear to add limited utility for distinguishing between classes.

In terms of class-specific patterns, Wave forms a distinct cluster in PC1 vs. PC2, indicating that these components capture its repetitive movement patterns well. Clench also appears tightly clustered, particularly in PC1 vs. PC2 and PC1 vs. PC3, which may be attributed to its consistent intensity and well-defined signal features. Rub demonstrates a prominent cluster in plots involving PC2 and PC3, suggesting that these components are effective in capturing its unique movement characteristics. However, PC4 seems to contribute minimally to class differentiation, as seen in the scattered and less structured patterns of plots like PC1 vs. PC4 and PC3 vs. PC4.

Overall, the pre-stroke analysis highlights the relevance of PC1 and PC2 in separating classes like Wave and Clench, while PC3 and PC4 appear less critical due to their weaker separability and higher noise levels. This analysis reinforces the importance of focusing on higher-variance components to gain meaningful insights from the pre-stroke data.

3D Prestroke Scatter Plots



Analysis and interpretation:

These 3D scatter plots depict the relationships between the pre-stroke data's three principal components (PC1, PC2, PC3, and PC4). The points in the plots are to represent different classes: Rest, Wave, Clench, and Rub. These visualizations allow for a deeper analysis of how the chosen components interact and contribute to distinguishing between these classes.

The Wave and Clench classes form compact and distinct clusters in the 3D space, particularly in plots involving PC1. This indicates that these classes exhibit consistent patterns, which are effectively captured by the variance explained by PC1 and PC2. Similarly, the Rub activity shows some clustering, particularly in the PC1-PC2-PC3 plot, suggesting that its unique movement patterns are represented well by these components. In contrast, the Rest class is widely dispersed and overlaps significantly with other classes across all plots. This scattered distribution reflects variability or noise in the Rest activity, which likely lacks distinguishing features in the pre-stroke data.

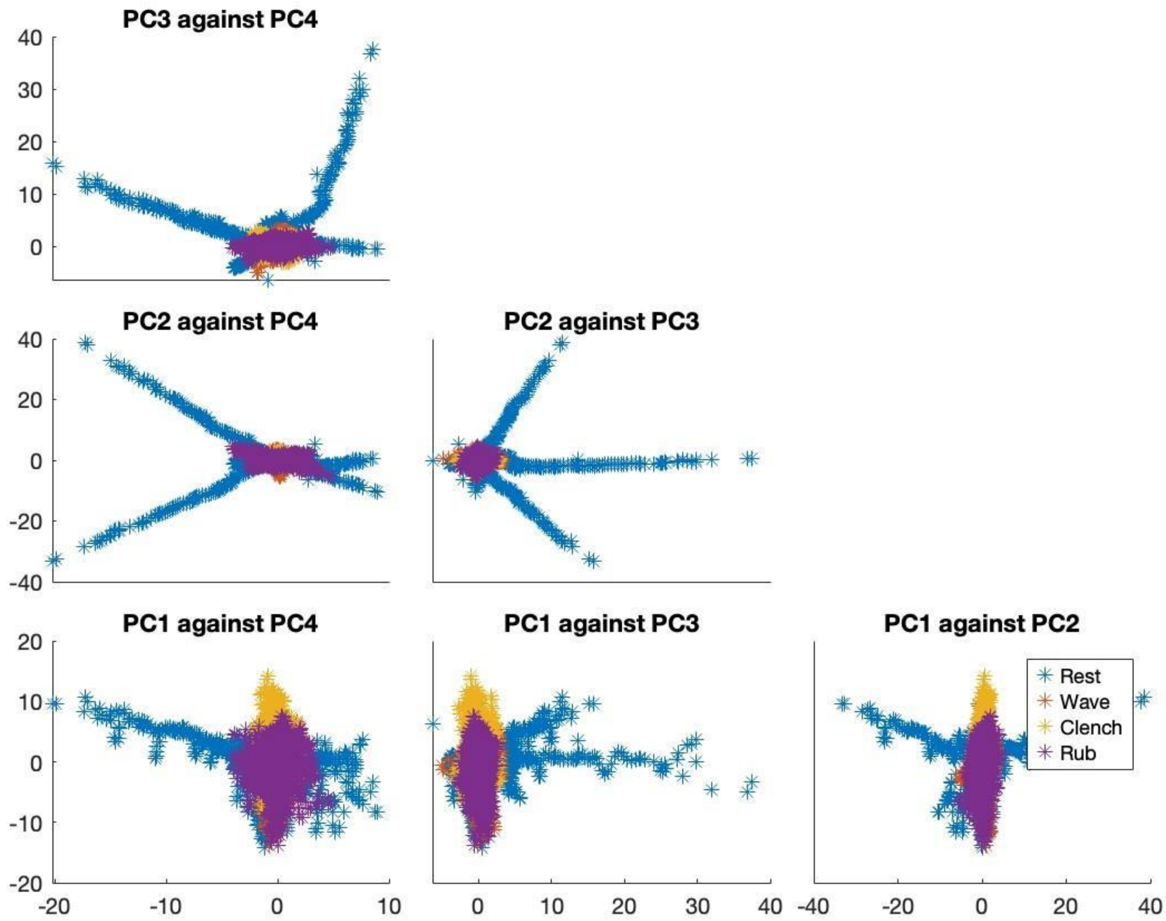
PC1 demonstrates its dominance in these plots, as evident by its wide range of values, particularly in PC1-PC2-PC4 and PC1-PC3-PC4. Its strong influence is apparent in how it helps separate classes like Clench and Wave. This indicates that PC1 captures the most significant and consistent patterns in the pre-stroke data, potentially reflecting key movement-related features or signal intensities.

The PC1-PC2-PC3 plot shows data points aligned along specific axes or diagonals, indicating a moderate correlation between PC1 and PC2. These two components likely capture shared features that are important for class differentiation. In contrast, plots involving PC4, such as PC1-PC2-PC4, show weaker and less organized relationships, suggesting a limited correlation between PC4 and the other components.

The analysis highlights that PC1 and PC2 are the most relevant components for separating the classes in the pre-stroke data. These components effectively capture distinct features for Wave, Clench, and Rub, reflecting dominant signal characteristics. Meanwhile, PC3 and especially PC4 appear less useful, as they contribute minimally to class differentiation and are more associated with noise or redundant features. The Rest class's scattered distribution underscores the challenge of distinguishing passive or baseline activity from more structured activities like Wave or Clench. This analysis suggests focusing on components like PC1 and PC2 for future classification or visualization tasks while considering dimensionality reduction to eliminate less informative components like PC4.

Poststroke data

2D Poststroke Scatter Plots :



ANALYSIS:

Plot 1: PC3 vs PC4

In the PC3 vs PC4 scatter plot, the resting state (blue) displays widespread scattering, indicating significant variability in neural signals post-stroke during inactivity. Motor tasks such as hand wave (orange), hand clench (yellow), and face rub (purple) form more compact clusters, reflecting task-specific neural engagement. The overlap between hand wave and face rub suggests shared activation of frontal regions (F3, F4) and central motor areas (C3, C4), which are critical for motor and sensory integration. The distinct clustering of hand clench emphasizes focused activation of central motor regions and parietal channels for precise hand coordination.

Plot 2: PC2 vs PC4

The PC2 vs PC4 plot highlights moderate dispersion in the resting state (blue) along PC4, representing neural instability during rest. Task-specific activities cluster tightly, especially along PC2. Hand wave (orange) and face rub (purple) exhibit overlaps, reflecting shared neural pathways likely involving frontal lobe activation for motor control. Hand clench (yellow) forms a more distinct cluster, which may be attributed to stronger activation of central (C3, C4) and parietal (P3, P4) regions that govern grip strength and fine motor skills.

Plot 3: PC2 vs PC3

In the PC2 vs PC3 plot, the resting state (blue) shows high variability, signaling challenges in achieving neural stability post-stroke. Motor activities form distinct clusters, but hand wave (orange) and face rub (purple) overlap, suggesting shared motor control mechanisms in the frontal and central areas. Hand clench (yellow) appears more separable, indicating distinct neural activation patterns, likely involving parietal regions (P3, P4) in conjunction with motor areas for hand-specific movements.

Plot 4: PC1 vs PC4

The PC1 vs PC4 scatter plot demonstrates the widest spread for the resting state (blue), reinforcing significant variability in baseline neural activity post-stroke. Clusters for motor tasks are more compact, but hand wave (orange) and face rub (purple) overlap substantially, pointing to shared frontal lobe involvement in motor and sensory coordination. Hand clench (yellow) stands out distinctly, reflecting focused neural engagement in central (C3, C4) and parietal regions (P3, P4), crucial for grasping motions and precision.

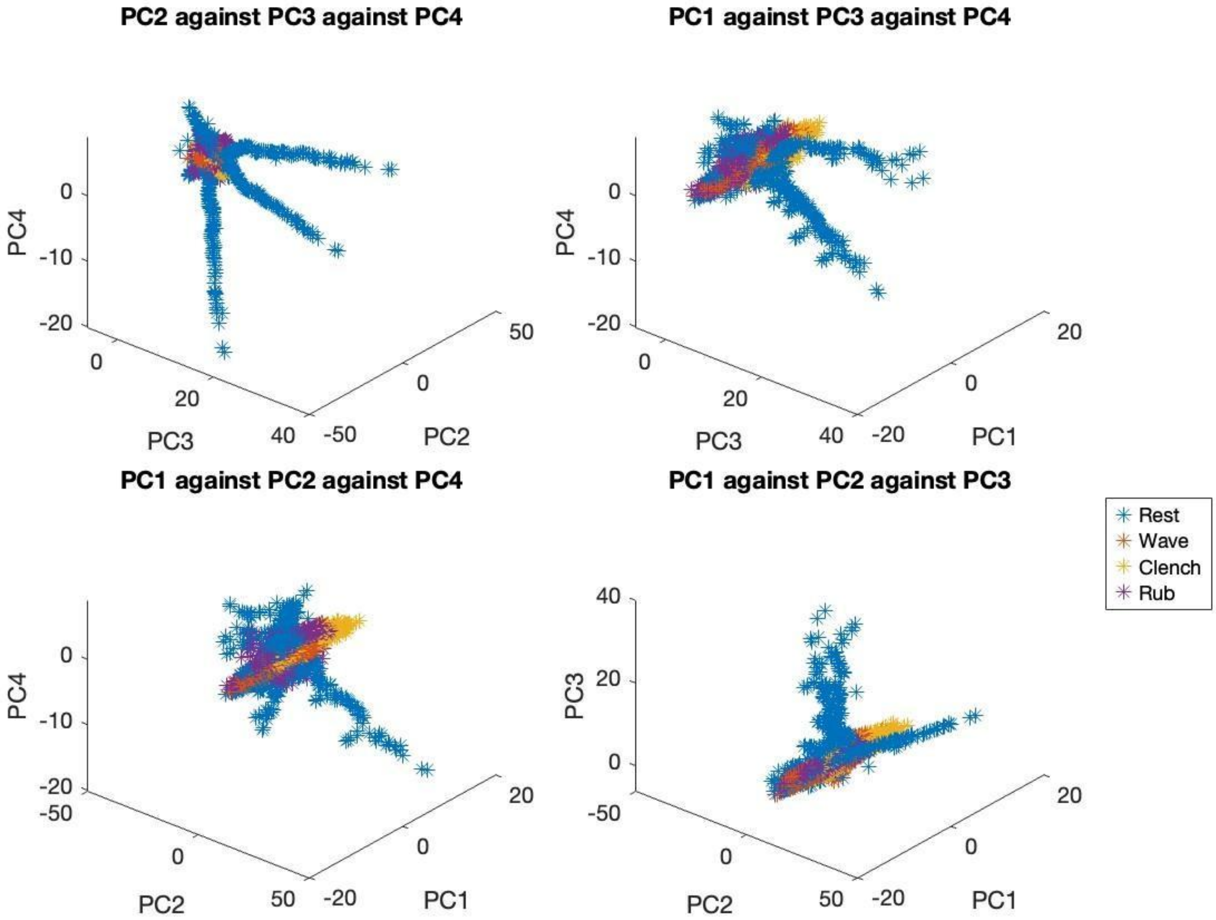
Plot 5: PC1 vs PC3

In the PC1 vs PC3 plot, resting state data (blue) is spread significantly along PC3, highlighting instability in neural signals during inactivity. Motor task clusters are evident, but overlaps between hand wave (orange) and face rub (purple) suggest similarities in frontal and central neural activation for these actions. Hand clench (yellow) remains relatively distinct, likely due to the involvement of central motor regions and parietal areas required for hand grip and strength coordination.

CONCLUSION:

These analyses highlight how stroke affects motor tasks by engaging specific brain regions. It reveals that post-stroke EEG signals show greater variability during rest and distinct clustering during motor tasks, with some overlap between similar activities. The frontal (F3, F4) and central (C3, C4) regions appear to play a pivotal role in coordinating these actions, with parietal regions (P3, P4) contributing to hand-specific tasks like clenching. The overlaps between tasks highlight shared compensatory neural pathways, while the spread in resting states reflects the broader impact of the stroke on baseline neural stability.

3D Poststroke Scatter Plots



ANALYSIS:

The data is visualized across principal components (PC1, PC2, PC3, and PC4), with motor tasks categorized into rest (blue), hand wave (orange), hand clench (yellow), and face rub (purple).

Plot 1: PC2 against PC3 against PC4

This plot illustrates the neural activity distribution for rest (blue), hand wave (orange), hand clench (yellow), and face rub (purple) across PC2, PC3, and PC4. The rest state forms a dense, centralized cluster, reflecting stable and consistent neural activity during inactivity. This stability suggests that the brain regions associated with passive states, such as the central (C3, C4) and parietal (P3, P4) channels, are minimally active, as no significant motor planning or execution is required. For hand wave, the data points are more dispersed, indicating moderate variability in the neural signals. This task engages frontal regions (F3, F4) for motor planning and the contralateral central channels (C3, C4) for execution. The dispersion suggests inconsistent coordination between these regions post-stroke, reflecting the subject's difficulty in maintaining a smooth motor rhythm. Hand clench exhibits a compact yet slightly dispersed cluster, implying relatively controlled neural activation. This task predominantly activates the motor cortex (C3, C4) and parietal regions (P3, P4), which are crucial for executing repetitive hand movements. The compact clustering reflects retained motor control, but the slight variability indicates impaired coordination in specific trials. Face rub, showing the broadest dispersion, suggests significant inconsistency in neural activity. This complex task involves the frontal (F3, F4) and central (C3, C4) regions for motor planning and execution, as well as sensory feedback from the parietal channels (P3, P4). The wide scatter reveals impaired integration between these regions, leading to inconsistent and uncoordinated movements.

Plot 2: PC1 against PC3 against PC4

In this plot, the rest state forms a dense, centralized cluster, indicating stable neural signals during inactivity. The lack of variance suggests minimal engagement of motor-related channels such as C3, C4, and frontal regions (F3, F4). Hand clench forms a relatively compact cluster with slight dispersion, suggesting focused yet somewhat variable neural activity. This task activates the central channels (C3, C4) for motor execution and the parietal regions (P3, P4) for spatial awareness and sensory integration. The compactness indicates that the subject retains a degree of motor control, but the slight dispersion reflects difficulty in maintaining consistent motor strength and rhythm. Hand wave, with its broader distribution, shows more variable neural activity. This action involves coordinated movement that heavily engages frontal regions (F3, F4) for motor planning and central channels (C3, C4) for execution. The variability suggests incomplete recovery of fine motor control, as these brain areas may struggle to consistently coordinate the motion. Face rub exhibits the widest dispersion, indicating the most inconsistent neural activity. This task involves complex coordination of the frontal, central, and parietal channels. The variability may result from impaired sensory feedback from the parietal regions and disrupted motor planning in the frontal cortex, causing erratic and uncoordinated movements.

Plot 3: PC1 against PC2 against PC4

This plot reveals clear separation between the rest state and active motor tasks. The rest state forms a dense and centralized cluster, reaffirming the stability of neural activity during inactivity, with minimal involvement of the motor and sensory brain regions. Hand clench appears as a distinct, relatively compact cluster, suggesting controlled neural activation. This task relies on the central channels (C3, C4) for motor execution and parietal regions (P3, P4) for sensory feedback. The compactness indicates retained motor control, while the small degree of dispersion suggests minor inconsistencies in execution. Hand wave shows moderate dispersion, reflecting variability in brain activation across trials. This motor action involves frontal (F3, F4) and central (C3, C4) regions, and the variability may arise from incomplete synchronization between motor planning and execution post-stroke. Face rub has the widest dispersion, with significant overlap with other tasks. This variability reflects the difficulty of this task, which requires extensive engagement of frontal, central, and parietal regions for motor planning, execution, and sensory integration. The overlap suggests that the brain channels struggle to activate distinct neural patterns for this action, resulting in inconsistent performance.

Plot 4: PC1 against PC2 against PC3

This plot highlights the stability of the rest state, with a tight and centralized cluster reflecting minimal brain activity. The central (C3, C4) and parietal (P3, P4) channels remain largely inactive during rest, resulting in consistent and stable EEG signals. Hand clench forms a relatively compact cluster, reflecting controlled and deliberate neural activity. The central channels (C3, C4) dominate during this task, while the parietal regions (P3, P4) provide sensory integration. The slight dispersion suggests occasional lapses in coordination due to impaired motor control post-stroke. Hand wave is more variable, with moderate dispersion. This motor task heavily engages the frontal (F3, F4) and central (C3, C4) channels. The variability reflects inconsistent motor planning and execution, as the neural circuits coordinating this movement may still be recovering. Face rub shows the most scattered distribution, indicating high variability in neural signals. This complex task involves significant engagement of the frontal (F3, F4), central (C3, C4), and parietal (P3, P4) channels. The wide scatter suggests impaired synchronization between these regions, leading to uncoordinated and inconsistent movements. The high variability reflects the difficulty of this task, which demands precise coordination and sensory feedback, both of which are likely affected post-stroke.

CONCLUSION:

Across all plots, the rest state exhibits the most stable and consistent neural activity, with minimal involvement of motor and sensory brain regions. Simple tasks like hand clench demonstrate relatively compact clusters, suggesting better-preserved motor control. However, more complex tasks like hand wave and face rub show broader and more scattered distributions, highlighting the challenges in coordinating motor planning (frontal channels), execution (central channels), and sensory feedback (parietal channels). These findings suggest that post-stroke rehabilitation should focus on improving coordination between these brain regions, gradually progressing from simple to complex motor tasks.

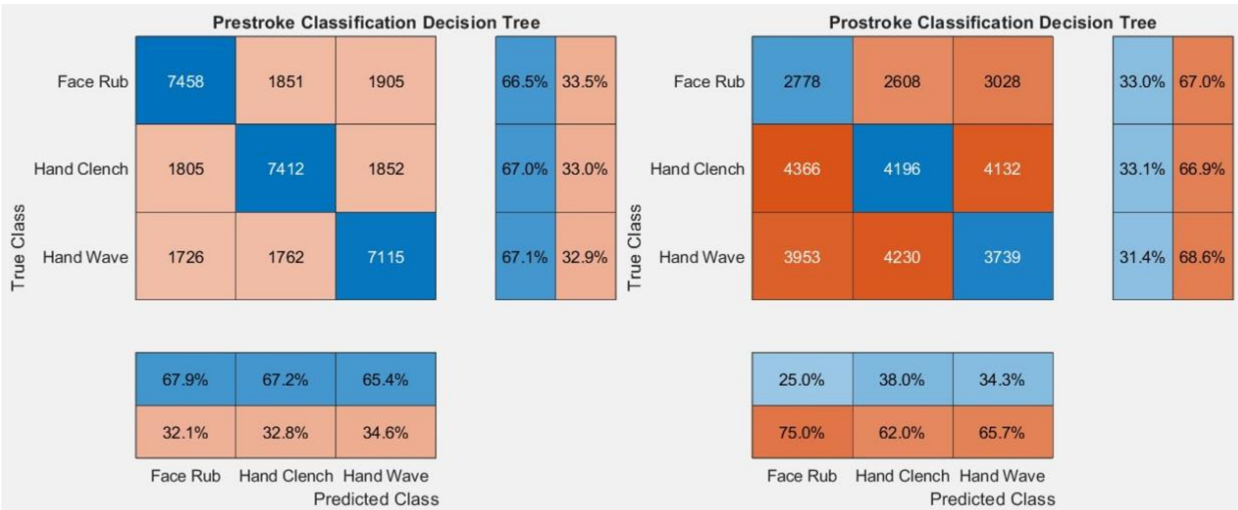
4.3 Classification Results

Two methods of analysis were performed on the dataset, having applied PCA processing. Supervised classification was used to find means of comparing the distribution of classes before and after the stroke event, whereas unsupervised classification was used to compare the stroke patient (Professor Mathews) subject's dataset with a non-stroke subject.

4.3.2 Supervised Classification

Results & Analysis

Decision Tree Classifier



These confusion matrices show the Prestroke and Poststroke classification results using a Decision Tree classifier. The confusion matrices are normalized and displayed side-by-side to allow comparison between pre-stroke and post-stroke performances.

Prestroke Classification Decision Tree:

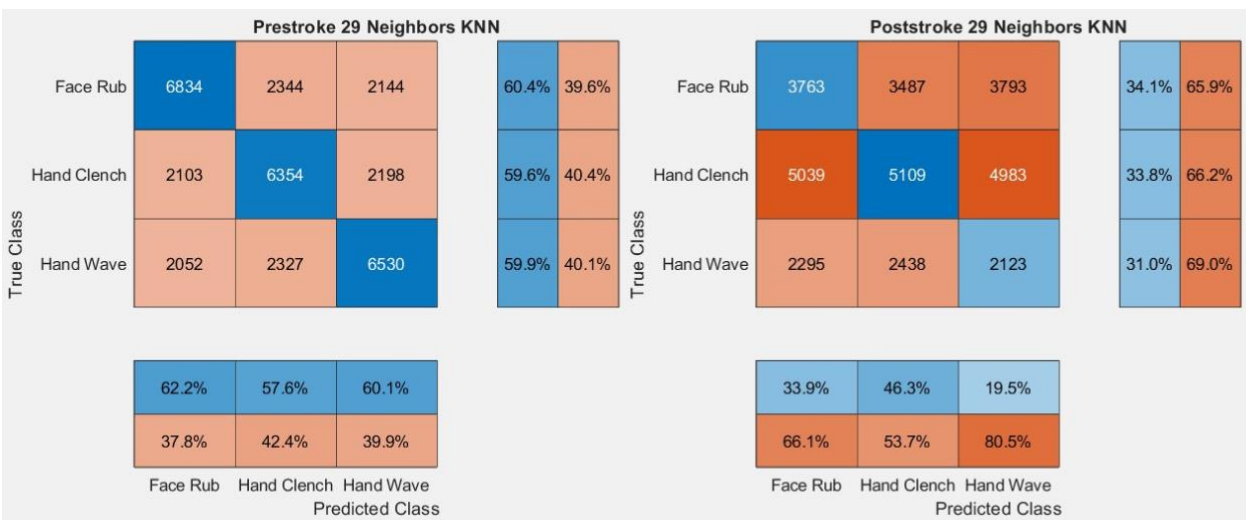
- The classifier is trained and tested on the prestroke dataset.
- The diagonal blue values represent correct classifications, while the off-diagonal orange values represent misclassifications.
- Face Rub: 67.9% of samples were correctly classified, with 32.1% misclassified into other actions.
- Hand Clench: 67.2% of samples were correctly classified, with 32.8% misclassified.
- Hand Wave: 65.4% of samples were correctly classified, while 34.6% were misclassified.
- Overall, the classifier demonstrates moderate performance, achieving roughly 65-68% accuracy for all three classes, with about one-third of the data misclassified.

Poststroke Classification Decision Tree:

- The classifier is trained on the prestroke data but tested on the poststroke dataset to evaluate generalization.
- Classification performance drops significantly compared to the prestroke results.
- Face Rub: Only 33.0% were correctly classified, with 67.0% misclassified into other actions.
- Hand Clench: 33.1% were correctly classified, with 66.9% misclassified.
- Hand Wave: 31.4% were correctly classified, with 68.6% misclassified.
- The results indicate poor generalization to poststroke data, with accuracy hovering around 31-33%, suggesting significant differences in the patterns between prestroke and poststroke movements.

The pre-stroke classifier shows good performance when trained and tested on the same dataset; however, it has difficulty generalizing to post-stroke data. This indicates a significant difference between pre-stroke and post-stroke patterns, emphasizing the challenge of predicting post-stroke movements using classifiers trained on pre-stroke data.

K-Nearest Neighbors Classifier



These confusion matrices show for the Prestroke and Poststroke 29 Neighbors KNN classifier, evaluating its performance on both datasets. Each confusion matrix is row-normalized to display the classification percentages, facilitating a clear comparison between correct and misclassified results.

Prestroke KNN:

- The classifier is trained and tested on the prestroke dataset.

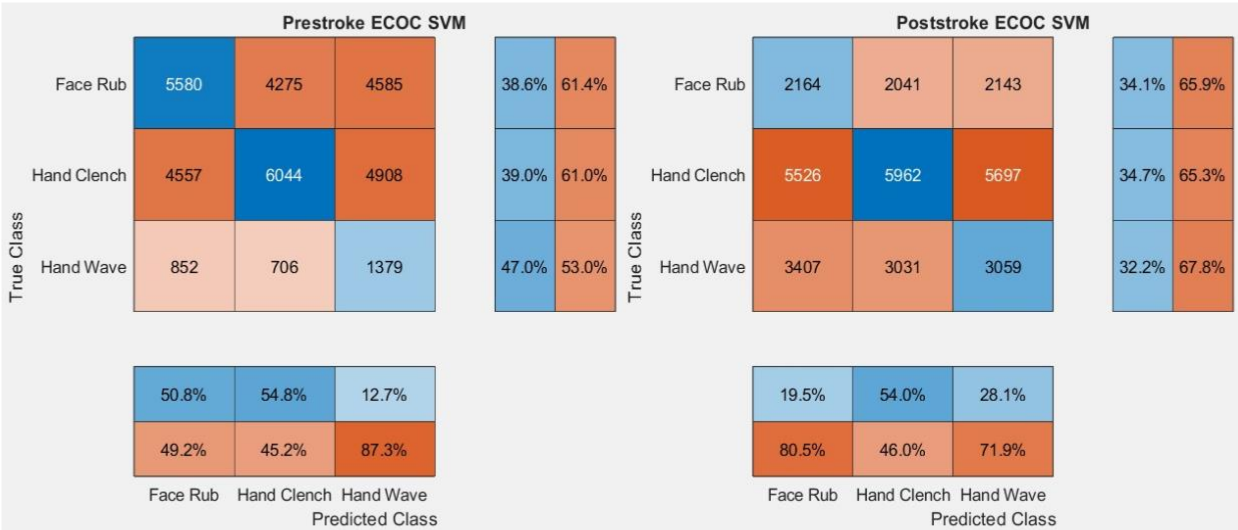
- Face Rub: 60.4% of samples were correctly classified, while 39.6% were misclassified into other actions.
- Hand Clench: 59.6% of samples were correctly classified, with 40.4% misclassified.
- Hand Wave: 59.9% of samples were correctly classified, and 40.1% were misclassified.
- Overall, the classifier achieves 59-60% accuracy across all three classes, with a considerable portion of 40% of data being misclassified, indicating moderate performance.

Poststroke KNN:

- The classifier, trained on the prestroke dataset, is tested on the poststroke dataset to assess generalization.
- Classification performance declines when tested on the poststroke data:
- Face Rub: Only 34.1% of samples were correctly classified, while 65.9% were misclassified.
- Hand Clench: 33.8% of samples were correctly classified, with 66.2% misclassified.
- Hand Wave: 31.0% were correctly classified, with 69.0% misclassified.
- The results reveal significant misclassification rates, with the correct classification percentages dropping to approximately 31-34% for all classes. This suggests poor generalization to poststroke data.

The KNN classifier shows moderate performance on the prestroke dataset, achieving an accuracy of approximately 59-60%. However, its performance significantly declines when tested on poststroke data, with accuracy dropping to around 31-34%.

ECOC Multiclass SVM:



These confusion matrices show for the Prestroke and Poststroke ECOC Multiclass SVM classifier. The results include the row-normalized classification accuracy percentages to better visualize the model's performance.

Prestroke ECOC SVM:

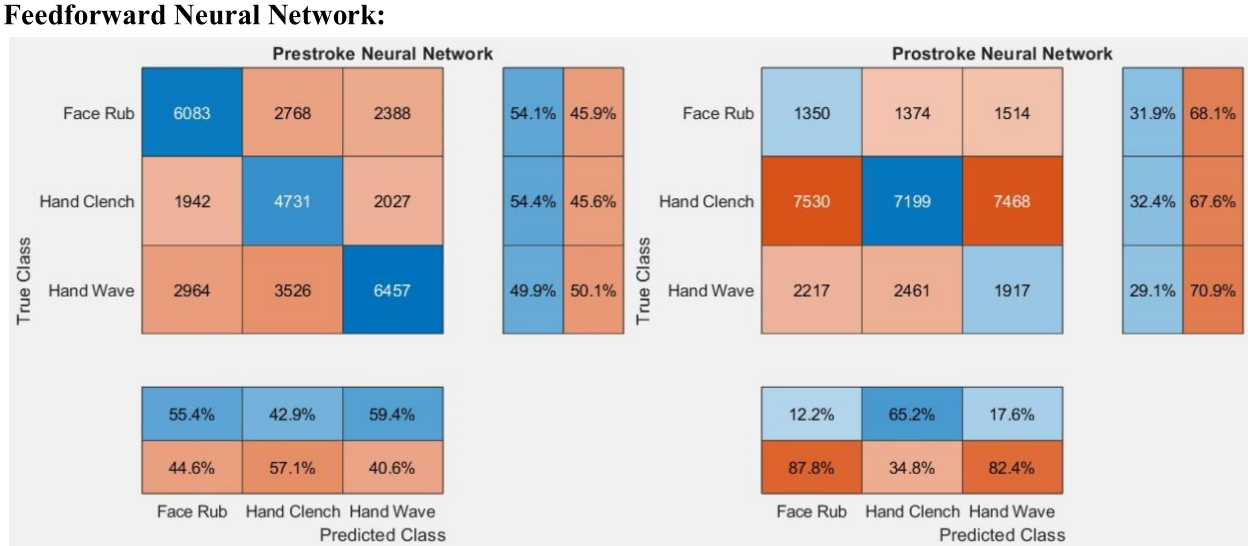
- The classifier is trained and tested on the prestroke dataset.
- Face Rub: 38.6% of samples were correctly classified, while 61.4% were misclassified as either Hand Clench or Hand Wave.
- Hand Clench: 39.0% of samples were correctly classified, with 61.0% misclassified into other classes.
- Hand Wave: 47.0% of samples were correctly classified, and 53.0% were misclassified.
- The column percentages at the bottom indicate the precision across predicted classes:
  - Face Rub: 50.8% precision.
  - Hand Clench: 54.8% precision.
  - Hand Wave: 12.7% precision.
- Overall, the prestroke results show the Hand Wave class suffers the most misclassification, with only 47.0% accuracy, and a significant percentage misclassified into other classes.

Poststroke ECOC SVM:

- The classifier, trained on the prestroke data, is evaluated on the poststroke dataset.
- Face Rub: 34.1% were correctly classified, with 65.9% misclassified.
- Hand Clench: 34.7% were correctly classified, and 65.3% were misclassified.
- Hand Wave: 32.2% of samples were correctly classified, while 67.8% were misclassified.
- Column precision scores for predicted classes:
    - Face Rub: 19.5% precision.
    - Hand Clench: 54.0% precision.
    - Hand Wave: 28.1% precision.
- Overall, the classifier performs worse on the poststroke dataset, with correct classification rates dropping to around 32-35% for all three classes. Misclassifications dominate, with most samples being incorrectly assigned to other categories.

The ECOC Multiclass SVM performs moderately well on the prestroke dataset, with classification accuracy ranging from 38% to 47% across the three classes. However, the model struggles to generalize to the poststroke dataset, where accuracy drops to between 32% and 35%, accompanied by significant misclassification rates of approximately 65% to 68%. Precision scores indicate that the Hand Clench class achieves the best classification performance compared to the other classes, particularly in the prestroke scenario.

Feedforward Neural Network:



These confusion matrices show the performance of the feedforward neural network classifier across pre-stroke and post-stroke datasets. The comparison is presented using confusion matrices for both datasets.

Prestroke Neural Network:

- The confusion matrix illustrates the classifier's performance when trained and tested on the pre-stroke dataset.
- Each row represents the true class labels (Face Rub, Hand Clench, and Hand Wave), while the columns represent the predicted class labels.
- Face Rub has a balanced split between correct predictions of 54.1% and misclassifications.
- Hand Clench achieves a correct prediction rate of 54.4%.
- Hand Wave is correctly predicted at a lower rate of 49.9%, with significant misclassification.

Prostroke Neural Network:

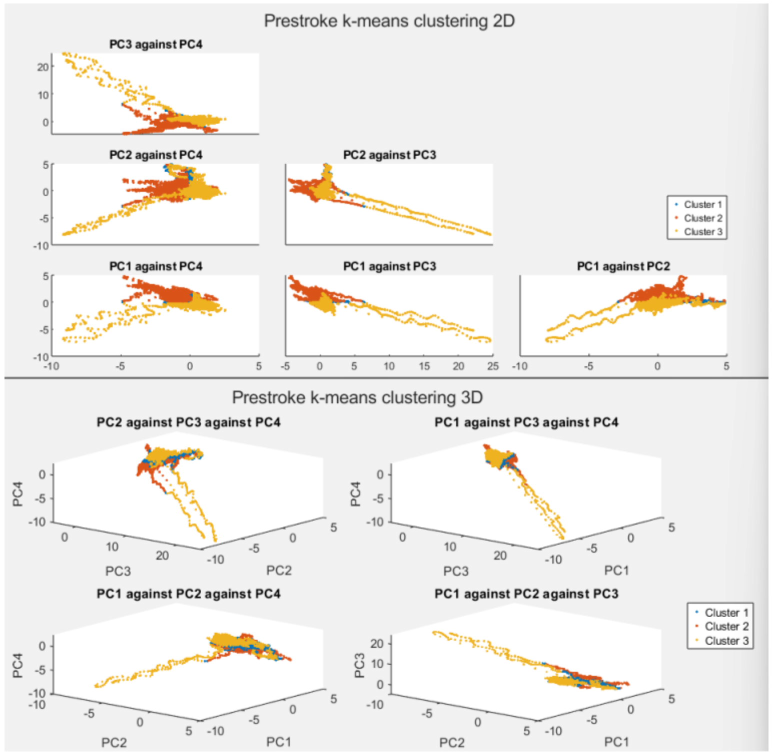
- The classifier's generalization is tested on the post-stroke dataset using the pre-stroke model.
- Misclassifications are more prevalent compared to pre-stroke performance.
- Face Rub is correctly classified 31.9% of the time, indicating a significant drop in performance.
- Hand Clench remains the best-performing class with 32.4% accuracy.
- Hand Wave drops further, with only 29.1% correct predictions.

The pre-stroke classifier exhibits an overall stronger performance compared to the post-stroke results, which reveal the challenges of classifying movement patterns after a stroke. The feedforward neural network shows a significant decline in performance when moving from pre-stroke to post-stroke data, underscoring the difficulties in detecting movement patterns following a stroke. These findings indicate the necessity of fine-tuning the model or exploring alternative methods for post-stroke classification.

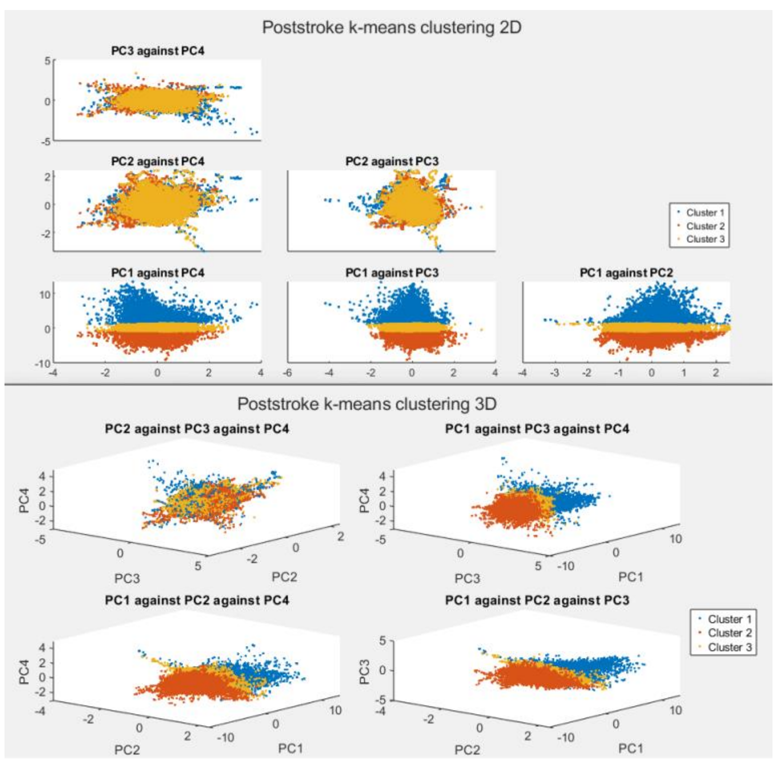
4.3.3 Unsupervised Classification

Results

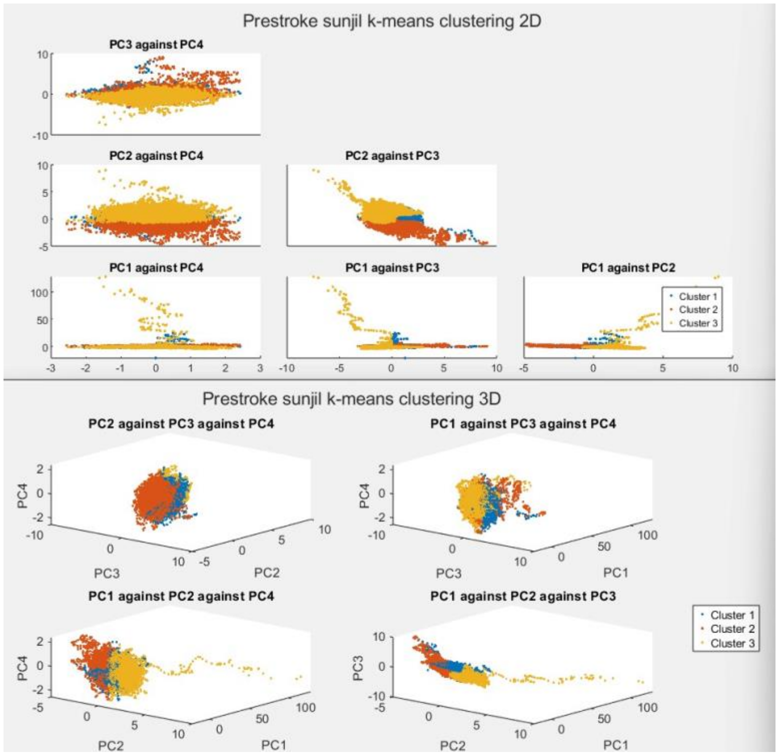
Professor Mathew Prestroke:



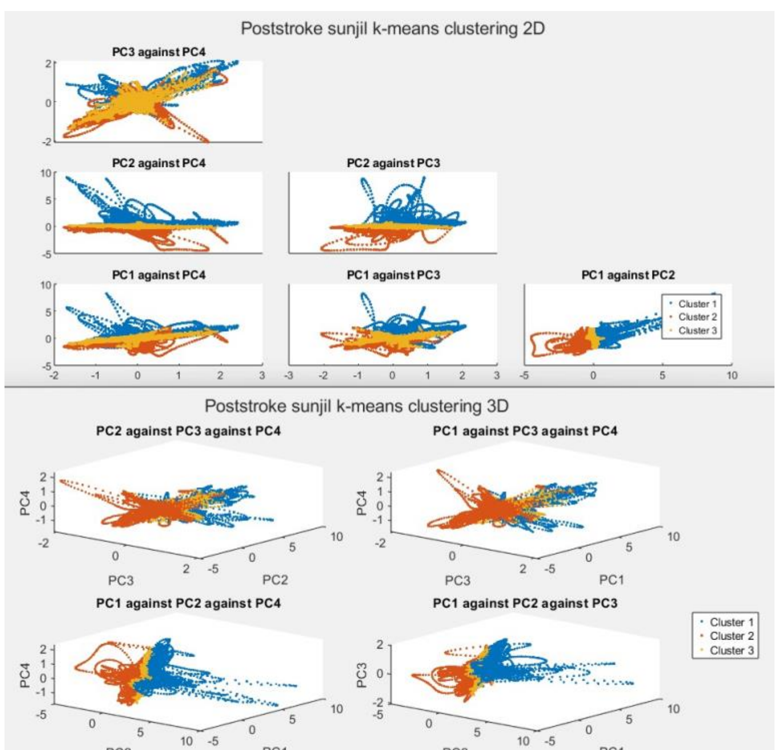
Professor Mathew Poststroke:



Sunjil Prestroke:



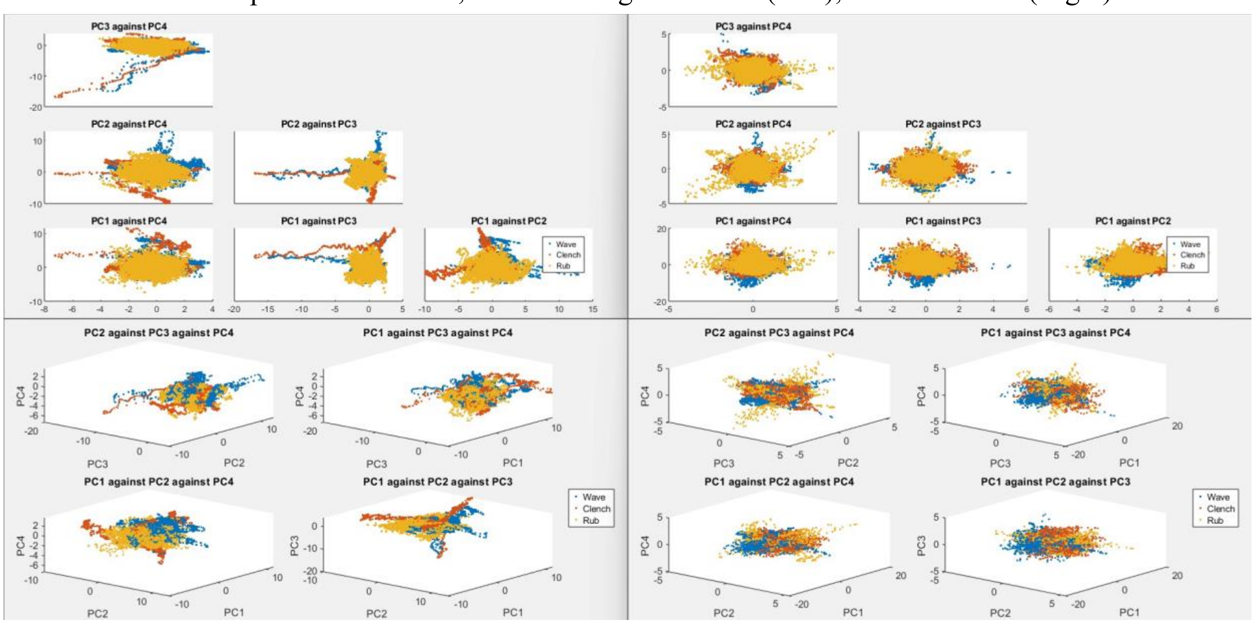
Sunjil Poststroke:



Analysis

Judging the formation of clusters overall, it can be observed in both 2D and 3D that clear delineations are present across the "space" of information. While each cluster does not correlate very closely to the true class plots of each dataset, the patterns visible in the produced clusters in relation between pre & post stroke are the most relevant comparisons.

Pictured: True class plots of PCA data, not clustering. Prestroke (Left), and Poststroke (Right)



Analysing Professor Mathew's data first, in terms of observable patterns, it can be observed that the "line artifacts", previously identified during the initial PCA analysis, still persist and are clearly divided across the clusters. The fact that these lines persist even after the block averaging and clustering makes them significant patterns, despite not perfectly lining up with the actual labels. These lines and the clustering thereof do not continue into the post-stroke dataset, meaning these same patterns are not reflected in the data. In contrast, line artifacts were found much more prominently in both pre & post-stroke time periods for Sunjil's dataset. This behaviour being present in a stroke patient but not in a non-stroke patient supports the hypothesis that these line patterns are a sign of normal brain functioning, and the lack of them in post-stroke data is a sign of degradation.

Another pattern revealed by the clustering results is the much more variable sizes of clusters in pre-stroke compared to post-stroke dataset. The Cluster 1 in Professor Mathew's pre-stroke represents a much smaller set of datapoints compared to the other two clusters which are relatively evenly split. In comparison, the post-stroke dataset shows a much more evenly distributed size of each cluster, with plots PC1-PC3-PC4, PC1-PC2-PC4, and PC1-PC2-PC3 displaying near identical clustering patterns. This could be representative of the post-stroke data being far more homogenous, and displaying less disparity between classes. However, similar behviou can also be seen the Sunjil datasets, with sunjil's prestroke dataset having a much diminished Cluster 3, vs post-stroke being more evenly distributed and identifiable. This could mean this behviour is instead a sign of the EEG devices degradation, and perhaps should not be taken into account.

In terms of clustering ability alone, the more variable nature of the post-stroke class data compared to the pre-stroke was only captured by the clustering in a minor way, being prominent only in the PC2-PC3-PC4 clustering plot for Professor Mathew. Otherwise, clusters formed neater, well defined cross sections of the plotted data. This likely points out the limited application of k-means clustering in this context, as it appears unable to account for high variability and class overlap.

One factor making drift even harder to establish was the differences in actions recorded for each dataset. For Professor Matthew's pre- & post-stroke, and Sunjil's pre-stroke, the action flow recorded was Rest -> Hand Wave -> Rest -> Hand Clench -> Rest -> Face Rub -> Rest, and this action-flow is what the entirety of the research is based upon. However, recordings of this action set were not available for Sunjil's post-stroke recording session. In the end, we made a concession to use one that had mostly analogous actions, such as "Fist", "Wave In" & "Wave out", and discarding data for actions that did not fit in with our research. This reduces the applicability of this analysis for comparing differences in individual actions.

CHAPTER 5: CONCLUSION

5.1 Conclusion

The PCA results revealed that, while the pre-stroke data showed clear separation between motor tasks, the post-stroke data exhibited more overlap between tasks.his suggests that, post-stroke, the brain's motor functions become less distinct and more generalized, which may contribute to the difficulty in accurately classifying motor tasks using classifiers trained on pre-stroke data.

The issue of **transitional states** between motor tasks was another significant finding. Motor tasks like hand wave, hand clench, and face rub, though distinct, share overlapping neural pathways, particularly during the transitions between one task and another. In the post-stroke data, these transitional states were more pronounced, with significant overlap in the neural signals corresponding to different motor tasks. This overlap likely contributed to the misclassifications observed in all classifiers. The inability of the models to effectively separate motor tasks during transitions indicates that post-stroke motor recovery involves more fluid and less defined shifts between motor states, which presents a challenge for accurate classification.

This study also investigated the ability of various machine learning classifiers—Decision Tree, K-Nearest Neighbors (KNN), ECOC Multiclass Support Vector Machine (SVM), and Feedforward Neural Network—to classify motor tasks (Face Rub, Hand Clench, and Hand Wave) before and after a stroke. The results highlight significant challenges in generalizing pre-stroke data to post-stroke scenarios, underscoring the differences in motor patterns and neural activity before and after a stroke.

For the **Decision Tree classifier**, the results were promising for pre-stroke classification, with accuracy ranging from 65% to 68% for all motor tasks. However, when the classifier was tested on post-stroke data, performance significantly declined, with accuracy dropping to around 31% to 33%. This stark difference indicates that the patterns learned from pre-stroke data are not generalizable to post-stroke data, revealing the challenge of using pre-stroke classifiers in predicting post-stroke movements.

The **K-Nearest Neighbors (KNN) classifier** showed similar trends, with moderate performance (59-60% accuracy) on the pre-stroke dataset. However, its generalization to post-stroke data was poor, with accuracy dropping to approximately 31-34%. These results further emphasize the difficulty in transferring motor task classification models from pre-stroke to post-stroke datasets.

The **ECOC Multiclass SVM** demonstrated a moderate performance on the pre-stroke dataset, achieving accuracy in the range of 38% to 47%. However, post-stroke classification accuracy again showed a significant decline, with accuracy falling to 32-35%. Precision scores highlighted that **Hand Clench** performed the best, though all classes suffered from high misclassification rates, particularly in post-stroke data.

The **Feedforward Neural Network** classifier exhibited similar trends, with moderate performance on the pre-stroke dataset but poor generalization to post-stroke data. The classifier's accuracy for **Face Rub** and **Hand Wave** dropped significantly, further confirming the challenges in detecting post-stroke movement patterns.

In conclusion, while all classifiers showed reasonable performance in the pre-stroke data, their ability to generalize to post-stroke movements was limited, suggesting a significant shift in motor patterns due to stroke. These findings indicate the necessity for further research to refine classifiers or explore alternative methods for stroke rehabilitation and motor function prediction. This may include collecting more diverse post-stroke data or developing models that can better adapt to the dynamic nature of stroke recovery.

5.2 Future Work

This study provides valuable insights into the use of EEG data for stroke rehabilitation, particularly in understanding motor function recovery and the role of different brain regions in the process. However, several aspects can be expanded upon in future research to further improve the understanding and applicability of EEG in stroke rehabilitation.

1. **Expanding the Dataset:** One of the main limitations of this study is the relatively small sample size, as the dataset primarily consists of EEG data from one subject (Professor Matthew) and one control subject (Sunjil). Future work would benefit from including a larger and more diverse sample of stroke patients, covering a range of ages, severity levels, and rehabilitation progress. This would help improve the generalizability of the findings and allow for the identification of more robust, population-wide trends.
2. **Incorporating Longitudinal Data:** This study analyzed pre- and post-stroke EEG data but did not include longitudinal data, which would allow for a better understanding of how motor function recovery evolves over time. Future research could involve collecting EEG data at multiple time points throughout the rehabilitation process, enabling a more comprehensive analysis of neural recovery and the factors influencing it.
3. **Integration with Other Physiological Signals:** Although this research focused solely on EEG, future work could explore the integration of EEG with other physiological signals, such as Electromyography (EMG), functional Near-Infrared Spectroscopy (fNIRS), or even wearable motion sensors. This multimodal approach could provide a more holistic understanding of motor function and recovery, combining brain activity, muscle activation, and movement patterns for more accurate predictions.
4. **Development of Real-Time Feedback Systems:** A promising direction for future research is the development of real-time EEG-based feedback systems that can assist stroke patients during rehabilitation. By providing continuous feedback on motor task performance, these systems could

help patients optimize their rehabilitation exercises and engage more actively in their recovery process. This would also require the integration of real-time machine learning models that can update predictions based on continuous EEG data.

5. **Exploring Recovery Patterns in More Complex Motor Tasks:** The focus of this study was on simple motor tasks such as hand wave, hand clench, and face rub. Future work could extend this analysis to more complex motor tasks, including tasks that involve bimanual coordination or fine motor skills. This would provide a deeper understanding of how different brain regions work together during more intricate movements and how recovery varies across different motor abilities.

In conclusion, while this study makes significant strides in understanding the role of EEG in stroke rehabilitation, there are numerous avenues for further research that could enhance our ability to monitor and predict recovery, providing a more personalized and effective approach to rehabilitation.